

Rollout-Aware Coefficient Activity for Data Selection in Verifiable-Reward RL

Anonymous authors

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Abstract

RL with verifiable rewards often treats task curricula and failure recycling as estimator-agnostic data interventions. We ask a narrower question: what task-level activity does the deployed group estimator make available to either intervention? For the zero-stabilizer form of practical MaxRL with N i.i.d. binary rollouts, the expected absolute coefficient mass is exactly $2(\text{pass}@N - \text{pass}@1)$. This rollout-aware activity score vanishes at pass rates 0 and 1 and has a unique, compute-dependent interior peak. It is an estimator-side diagnostic, not a theorem of learning progress. In a balanced neural maze factorial, an external execution record reports that the time-integrated MaxRL-GRPO coverage contrast was fixed before a six-block fresh wave; it was positive in 6/6 blocks under each sampler ($p=.031$ per sampler). Averaging the correlated sampler contrasts within seed gives a wave-2 mean of $+.0195$ with a post-hoc 95% interval $[+.0115, +.0275]$. On Countdown, a reported three-seed recycling-package aggregate raises mean@16 from .278 to .324 while the logged VERL bootstrap best@16 proxy falls from .541 to .492; this proxy is not standard unbiased pass@16, and missing raw task outcomes prevent recomputation. A higher-dose replay arm improves both logged metrics but does not isolate relabel direction; two matched controls failed treatment delivery, and the proposed saturation gate did not validate under corrected code. Together, these results support a calibrated conclusion: curricula and recyclers must be evaluated jointly with the estimator beneath them, and mean accuracy alone can hide a loss of coverage. We also show that dropping all-fail groups targets truncation order $T=N-1$, not N — a property of the deployed centered-and-dropped convention, not a revision of MaxRL’s theorem for its uncentered conditional estimator.

Status. This is the extended research record: every rung, every retraction, and the full implementation detail. The *current scope of claims* is the 9-page ICLR submission body (`paper/body_iclr.tex`), which is narrower — it treats coefficient activity as an estimator-conditioned source of curriculum hypotheses rather than a curriculum law, and reports the two preregistered boundaries in §8. Where the two differ, the submission body governs.

1 Introduction

Post-training language models and control policies with verifiable rewards have a cost structure unlike supervised learning: the data are free, but *rollouts* are not. Every optimization step is preceded by sampling N completions per task, and generation can dominate wall-clock — in our committed telemetry, generation-dominated LLM sessions average 83% of step time.¹ On hard task pools, most of that compute buys nothing: under uniform sampling we measure that **65–75% of rollout groups produce zero learning signal** — either every rollout fails (the group is dropped by success-conditioned estimators) or every rollout succeeds (no contrast, no gradient).

The field’s response is a pair of data-level interventions. The first is the *difficulty curriculum*: reallocate the rollout budget toward tasks where learning is possible right now — UCB bandits over difficulty buckets [8], target-difficulty controllers [9], category bandits [10], learnability-based rejection sampling [11], dead-prompt filtering [12, 14]. The second, newer, is *failure recycling*: convert a failed rollout into a verified success of the task it actually achieved — hindsight relabeling, recently brought into RLVR-style loops for robot policies

¹Eight sessions, 282 steps; across all 1,188 logged steps including CPU-bottlenecked maze-scale sessions the step-weighted mean is 31% — the 83% figure describes the LLM regime, not every workload.

[25]. Both interventions are, in current practice, designed and evaluated as if they were *objective-agnostic*: bolt-on modules whose benefit is assumed to transfer across the advantage estimator they sit on, and whose cost is assumed visible in the mean-accuracy metrics they are tuned against.

Neither assumption is safe without inspecting the estimator. Our organizing quantity is $u_N(p)$, half the estimator’s expected absolute coefficient mass. It vanishes at pass rates 0 and 1 and peaks once in between, separating a dead zone, a learnable region, and a mastered tail. Curricula reallocate compute within the learnable region; verified relabeling can create an auxiliary update for an all-fail request; and relabeling can concentrate updates on goals the policy already reaches. The estimator shapes the activity available in all three cases, while score geometry and parameter sharing determine what that activity ultimately does. Three questions organize the paper:

Q1: What can a curriculum contribute, at most? For success-conditioned advantages, the expected coefficient mass a task emits from a group of N rollouts has a closed form: $2(\text{pass}@N - \text{pass}@1)$, a compute-indexed learnability functional whose two exact zeros anchor the difficulty regimes (§3). Mass is a surrogate for learning signal (Remark 1), but it is the part a sampler can act on before spending a rollout, and it makes the allocation *derivable* rather than heuristic: the learnable region’s location, peak, and N -scaling come from the budget you already chose, and the common $p(1 - p)$ learnability score is the $N=2$ slice. It also bounds the allocation channel. A sampler can redistribute only activity that already exists. On the exact testbed, a matched perfect-information allocator and the posterior-plus-recycling stack finish within noise, while recycling adds only a small increment on top of that oracle (§6.1). At LLM scale, a prompt-level posterior can also be too data-starved to deliver meaningful steering (§6).

Q2: When is either intervention safe? The same algebra says the answer depends on the estimator. MaxRL concentrates $(N-1)\times$ more mass than RLOO (the unnormalized leave-one-out baseline estimator) on hard tasks as $p \rightarrow 0$ and releases mastered tasks; GRPO’s variance-normalized weights keep spending on both tails. The predicted coverage ordering is robust where gradients are exact; at neural scale our first pre-registered factorial *retired its strong form* — an earlier cohort’s zero-exception divergence had conflated recycling’s contribution with the estimator effect — and a second factorial on fresh seed blocks then *confirmed* the time-integrated ordering that survived, 6/6 paired blocks under each sampler (§6.3b). Recycling has a separate risk: the achieved-goal distribution is tilted toward outputs the policy already reaches. On Countdown, the reported aggregate has higher mean accuracy and a lower logged bootstrap coverage proxy. This is consistent with concentration, but the missing raw outcomes and failed matched controls prevent a causal attribution. A higher-dose live-group replay arm improves both logged metrics, showing that generic extra optimization can explain at least part of the mean gain without identifying the effect of relabel direction (§6.8).

Q3: Does the algebra identify a reliable remedy? It suggests a testable heuristic, not a solved mechanism. We gate relabel admission using a recent-frequency statistic for achieved destinations, motivated by the mastered-tail zero of u_N . The favorable moderate setting ran under faulty decay; a corrected strong setting failed to reproduce a useful trade-off, so no gate operating point is validated. Jugs further shows that both the relabel map and destination key are domain-specific and load-bearing (§6.9 and §9). The failed remedy is part of the result: estimator algebra can locate a risk without by itself specifying a deployable controller.

Our contributions:

1. **A closed-form learnability functional (§3):** the expected coefficient mass of a prompt under success-conditioned RL is exactly $2(\text{pass}@N - \text{pass}@1)$, compute-indexed, with peak location and N -scaling derived rather than tuned, and exact zeros at $p \in \{0, 1\}$ anchoring the dead zone, the learnable region, and the mastered tail; the canonical binary-success $p(1 - p)$ learnability score is its $N=2$ slice. *Derived is not the same as optimal:* a later preregistered exponent sweep (§8) rejects the deployed- N peak *location* while supporting the harder-peaked shape. (The functional is closed-form; the sampler built on it still estimates p and keeps standard knobs.)
2. **A derived teacher and a characterized recycler (§4–5):** FrontierMax schedules prompts by this functional with the per-group estimator untouched, and hindsight relabeling under a verified-success contract is characterized as a verified, adaptively selected auxiliary update — on-policy for the achieved task only under an explicit distribution-matching condition (Remark 3).
3. **An estimator-conditioned coverage ordering, retired in its strong form and confirmed in its measured one (§6):** our first pre-registered factorial retired the cohort’s zero-exception divergence at its

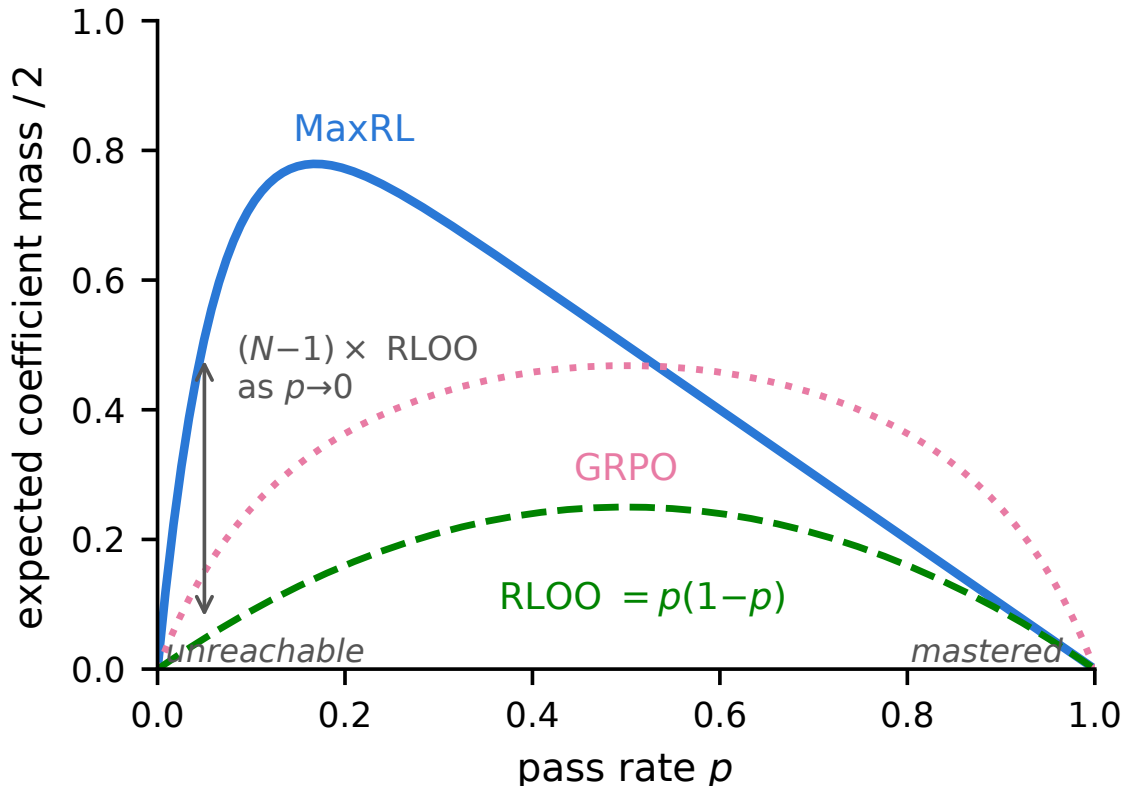


Figure 1: **The estimator is the curriculum.** *Left:* the derived utility $u_N(p) = (1 - (1 - p)^N) - p$ — half the estimator’s expected coefficient mass $A_N = 2u_N$ (Prop. 2) — for growing rollout budgets N . The peak $p^* \approx \ln N/N$ walks toward harder prompts as compute grows; the utility vanishes exactly at $p \in \{0, 1\}$. *Right:* all three curves are exact expected half-mass at $N=16$ for the estimators *as deployed* (GRPO with sample-SD normalization: $\sqrt{\frac{N-1}{N}} \frac{1}{N} \mathbb{E} \sqrt{K(N-K)}$, $K \sim \text{Bin}(N, p)$; MC-verified against the code). MaxRL concentrates $\rightarrow (N-1) \times$ more mass than RLOO on frontier prompts (Prop. 3); in the mastered tail MaxRL and RLOO decay together at first order (both $\rightarrow q = 1 - p$; MC artifact `verify_guidance_math.json`) while GRPO’s variance normalization keeps $\sim (N-1)/\sqrt{N} \times$ their relative mass there — the maintenance-of-easy-tasks its curriculum arms lose in §6. RLOO’s profile is exactly the published “learnability” objective — the $N=2$ slice.

registered endpoint; the time-integrated ordering that survived it was then registered as the primary of a second factorial on fresh seed blocks and confirmed 6/6 under both samplers ($p=.031$ each; positive in 12/12 independent blocks across both waves after within-block sampler averaging), with controls closing the scheduler-mismatch and variance-normalization readings (§6.3b).

4. **An applied metric warning and an unresolved causal gap** (§6.8–6.9): the reported three-seed Countdown aggregate combines higher mean accuracy with a lower logged bootstrap coverage proxy. A higher-dose replay control shows what generic extra optimization can buy but does not isolate relabel direction; two matched controls failed delivery. The proposed saturation gate also failed corrected-code validation. We therefore retain the metric trade-off and state the causal mechanism and mitigation as open.

Along the way we state a refinement to the MaxRL paper — the practical drop- $K=0$ estimator is unbiased for the $T=N-1$ objective, not $T=N$ (Lemma 1) — and every prediction below was committed before the deciding runs, with negatives and retractions reported in place (§9) and audited in the repository; App. D tabulates every load-bearing claim with its evidence and registration status.

2 Background: MaxRL in three lines

For binary rewards, maximum likelihood maximizes $\mathbb{E}[\log p_\theta(\text{success} \mid x)]$. MaxRL expands $\log p$ as a Maclaurin series in $(1 - p)$ and truncates at order T , giving a compute-indexed family $J^{(T)}$ interpolating REINFORCE ($T=1$) to exact ML ($T \rightarrow \infty$); the practical estimator sets per-rollout advantages $w_i = r_i/K - 1/N$ with all-fail groups dropped (Lemma 1 gives its exact truncation order). Its weight function grows as $p \rightarrow 0$: the objective is an *implicit, gradient-level* curriculum. Our work is the data-level complement: the same algebra, read as a sampling rule plus a recycling rule.

3 The estimator’s activity geometry

Proposition 1 (Coefficient mass under an arbitrary group law). *Under the practical centered drop-all-fail convention with zero stabilizer, the realized absolute coefficient mass of a group with K successes is the deterministic function $M(K) = 2(1 - K/N)\mathbf{1}\{K > 0\}$. Hence for any joint binary group law Q on $\{0, 1\}^N$ — no independence and no identical distribution assumed —*

$$A_N(Q) = \mathbb{E}_Q[M(K)] = 2 \left(\Pr_Q(K > 0) - \frac{\mathbb{E}_Q[K]}{N} \right) = 2(q_N - \bar{p}).$$

Specializing to $Q = \text{Bernoulli}(p)^{\otimes N}$ gives $q_N = 1 - (1 - p)^N$ and recovers $A_N(p) = 2\{1 - p - (1 - p)^N\}$.

Interpretation. *The coefficient-mass identity requires no independence at all. What requires conditionally i.i.d. rollouts is the scalarization: compressing the group law Q_x into the single number p_x , and with it the factorization $p(1 - p)w_{N-1}(p)$ and the peak p_N^* . Estimator activity remains exactly realized activity; what can fail is predicting it from a coarse mean pass rate.*

Corollary 1 (Task granularity). *Let a curriculum score an aggregate z (a level, template, or domain) whose concrete tasks are $X \sim \nu(\cdot \mid z)$, with rollouts conditionally i.i.d. given X . Since $u_N''(p) = -N(N - 1)(1 - p)^{N-2} < 0$, Jensen gives $\mathbb{E}_X[u_N(p_X)] \leq u_N(\bar{p}_z)$, and the gap is exact:*

$$A_N(\bar{p}_z) - 2\mathbb{E}_X[u_N(p_X)] = 2[\Pr(K=0 \mid z) - (1 - \bar{p}_z)^N] \geq 0,$$

twice the aggregate’s excess all-fail probability over the binomial at the same mean.

Interpretation. *Curvature decides who pays. Near u_N ’s own peak the penalty is $\approx \frac{1}{2}|u_N''(p_N^*)|\text{Var}(p_X)$ with $|u_N''(p_N^*)| = (N-1)/(1-p_N^*)$ — about 34.7 at $N = 32$ — while for the canonical score the relation is exact and far milder, $u_2(\bar{p}) - \mathbb{E}[u_2(p_X)] = \text{Var}(p_X)$. So raising N moves activity toward harder tasks and makes the geometry more sensitive to how coarsely the curriculum names a task. Both identities are verified to floating point on 288,000 real group draws in `curriculum_maxrl/maze_score/group_law_audit.py`; the contract tests in `curriculum_maxrl/test_group_law.py` exercise them against the deployed estimator under dependent, anti-correlated, heterogeneous and random dense joint laws.*

Lemma 1 (Truncation order of the practical estimator). *Let $J_T(p) = -\sum_{t=1}^T (1 - p)^t/t$ denote the T -truncated ML objective, so $\nabla J_T = (\sum_{j=0}^{T-1} (1 - p)^j)\nabla p$; $T \rightarrow \infty$ recovers exact ML and $T=1$ REINFORCE. Under N i.i.d. binary rewards with pass rate p , the drop- $K=0$ variant of the practical MaxRL estimator has expectation $(\frac{1-(1-p)^N}{p} - (1 - p)^{N-1})\nabla p = (\sum_{j=0}^{N-2} (1 - p)^j)\nabla p$: it is exactly unbiased for J_{N-1} , not J_N , for $N \geq 2$. Zeroing the control variate only on all-fail groups makes it outcome-dependent, contributing the $-(1 - p)^{N-1}\nabla p$ correction, and $\sum_{k=1}^N (1 - p)^{k-1} - (1 - p)^{N-1} = \sum_{k=1}^{N-1} (1 - p)^{k-1}$.*

Interpretation. *Three estimators must be kept distinct: the raw success-conditioned average ($\frac{1}{K}\sum r_i S_i$, no control variate — the object of Tajwar et al. [1]’s Theorem 2, exactly unbiased for J_N), the full control-variate form (subtracting the unconditional mean score on every group, including all-fail ones), and the practical drop- $K=0$ form above, which is what ships. The lemma is a refinement to Tajwar et al. [1], immaterial for their conclusions but load-bearing for our mass accounting at small N — all identities below are for the estimator as actually shipped. Magnitude at the deployed budgets: the relative gradient shortfall of $T=N-1$ vs. $T=N$ is $(1 - p)^{N-1}/\sum_{k=1}^N (1 - p)^{k-1}$ — at $N=16$ it is 5.8% at $p=.01$, 1.1% at the band peak $p^* \approx .17$, and $< 10^{-4}$ at $p=.5$; at $N=8$ it reaches 12% on the deep frontier. Bookkeeping at moderate p , material exactly where the frontier teacher concentrates sampling.*

Proposition 2 (Expected coefficient mass). *For a prompt with pass rate p and N i.i.d. rollouts under the practical MaxRL weights, define the utility $u_N(p) := \text{pass@N}(p) - \text{pass@1}(p) = (1 - (1 - p)^N) - p$. Then the expected absolute coefficient mass is*

$$A_N(p) := \mathbb{E} \left[\sum_i |w_i| \right] = 2(\text{pass@N}(p) - \text{pass@1}(p)) = 2u_N(p). \quad (1)$$

Proof (three lines): on $\{K \geq 1\}$, $\sum_i |w_i| = K(\frac{1}{K} - \frac{1}{N}) + \frac{N-K}{N} = 2(1 - \frac{K}{N})$, and $\mathbb{E}[2(1 - \frac{K}{N})\mathbf{1}\{K \geq 1\}] = 2(\Pr[K \geq 1] - p) = 2(\text{pass@N} - \text{pass@1})$. The sampling utility used everywhere below (figures, Algorithm 1, code) is u_N ; the factor two matters only when quoting exact mass, and we carry it explicitly as $A_N = 2u_N$. Two exact companions tie mass to the gradient. Factorization: writing $\mu_{\pm}(x) = \mathbb{E}[S \mid r=1/0, x]$ for the success/failure-conditioned score means, exchangeability gives, for every group size,

$$\mathbb{E}[\hat{g} \mid x] = u_N(p) (\mu_+(x) - \mu_-(x)), \quad (2)$$

so u_N is exactly the estimator-controlled scalar multiplying the prompt’s success–failure score contrast — the part of the expected update a sampler can know before rolling out; the contrast (and where it transfers) is the part it cannot. Weight view: since $\mu_+ - \mu_- = \nabla p/(p(1 - p))$, Eq. (2) is equivalent to $u_N(p) = p(1 - p)w_{N-1}(p)$ with $w_T(p) = (1 - (1 - p)^T)/p$ the truncated weight function of Lemma 1 — mass and the practical estimator’s weight view are linked, not identical objects. (Both identities MC-verified; artifact `verify_guidance_math.json`.)

Interpretation. *The estimator’s expected coefficient mass on a prompt is twice the probability that the prompt is solvable within N attempts but not within one. Coefficient mass is a surrogate for learning signal (Remark 1), but its two zeros are exact, and they anchor three regimes of difficulty. A dead zone at $p = 0$: nothing to condition on; sampling alone cannot produce a success there.² A learnable region with a unique interior peak at $p^* = 1 - N^{-1/(N-1)} \approx \ln N/N$ (unique because $u_N'' < 0$). A mastered tail at $p = 1$: no contrast left. The regimes are continuous — the only exact zeros are $p \in \{0, 1\}$ — so where we quote a “band” we mean the threshold set $\{p : u_N(p) \geq \eta u_N(p^*)\}$ ($\eta=0.5$ unless stated), whose location and N -scaling follow from the one parameter you already chose. Everything in this paper is one of three moves against this structure: allocate within the learnable region (§4), create signal at its bottom (§5), and refuse updates that land in the mastered tail (the gate).*

Proposition 3 (Frontier amplification). *As $p \rightarrow 0$, the ratio of MaxRL’s expected mass to RLOO’s ($A_N^{\text{RLOO}} = 2p(1 - p)$ exactly, under the $1/N$ -normalized leave-one-out weights) tends to $N - 1$.*

Interpretation. *The comparison to GRPO is two-sided and exact for the deployed implementation, which normalizes by the **sample** standard deviation ($s_K = \sqrt{K(N-K)/(N(N-1))}$, PyTorch’s default), not the population SD: $A_N^{\text{GRPO}} = 2\sqrt{\frac{N-1}{N}} \frac{1}{N} \mathbb{E}\sqrt{K(N-K)}$ with $K \sim \text{Bin}(N, p)$ (MC-verified against `estimators.py`). Taking tails, $\text{MaxRL/GRPO} \rightarrow \sqrt{N}$ as $p \rightarrow 0$ while $\text{GRPO/MaxRL} \rightarrow (N-1)/\sqrt{N}$ as $p \rightarrow 1$ (at $N=16$: $4.0\times$ and $3.75\times$; the symmetric $\sqrt{N-1}$ holds only under population-SD normalization). MaxRL over-serves the frontier and releases mastered prompts; GRPO silently keeps $\sim(N-1)/\sqrt{N}\times$ more of its mass on mastered prompts — maintenance a curriculum then removes. One caveat governs every cross-estimator magnitude in this paper, and we measured its bite: each estimator’s coefficients carry a conventional global scale (absorbable into the learning rate), so absolute ratios like these are facts about the implementations as shipped under a common learning rate, not invariant properties of the objectives — and on a learnable-everywhere pool an lr sweep shows a $2\text{--}4\times$ raise lets GRPO match MaxRL’s raw coverage delta (though not its coverage–reliability premium; §6.3b). What survives any recalibration is the shape: where each estimator’s mass concentrates as a function of p , and its exact zeros — which is why the empirical claims are about tails, signs, and premium-vs-reliability trades, not magnitudes. Concentrating sampling on the frontier only helps if the estimator extracts signal there, and §6 shows the mastered-tail difference becoming an active failure under a curriculum.*

The static known- p allocation problem this utility induces has a clean water-filling solution, but the deployed teacher does not use it (it keeps fixed group sizes and samples $\propto u^\gamma$ with a floor, trading myopic optimality for group-parallel batching and exploration); we state the allocator and its boundary conditions

²At small $p > 0$ a task is *operationally* dead at budget (B, N) when its live-group probability $1 - (1 - \ell_N(p))^B$, with $\ell_N(p) = 1 - p^N - (1 - p)^N$ and B its share of groups, is too small to sustain training. On the frontier-heavy pool of §6.2 ($p \leq 10^{-5}$, ~ 133 groups per task) that probability is $\leq 2\%$ per task, and the one live group observed across all uniform seeds did not ignite — dead in the budgeted sense, not by fiat.

in App. B and measure what the gap costs in §6.1, where a no-floor, γ -matched true-pass-rate oracle ties the Thompson teacher in end performance.

Remark 1 (Scope of the surrogate). Coefficient mass counts weighted rollouts, not score-vector geometry. We use it as the sampling utility because (i) it is knowable *before* spending rollouts, (ii) its zeros are exact — at $p \in \{0, 1\}$ the estimator emits no gradient of any norm — and (iii) on the exact-gradient testbed it *ties* the true first-order expected improvement as a task ranking (Spearman within .005 at every snapshot, and exactly 1.0 against the full exact transfer matrix — one expected group per task, exact Δp on every other task — at three training snapshots; artifacts `results_bridge.json`, `results_transfer_matrix.json`). That tie is a property of *shared-prefix* task structure, where activity and transfer align; on pools without it, activity ranks what a task can emit, not where it helps (§6.5–6.6 measure that gap). What it does not claim to be is *the* optimal sequential utility, and no state-local functional is: on independent tasks the true final-performance optimum is a planning solution (trajectory-knapsack with a budget-dependent funding cutoff), and which surrogate best approximates it depends on the training objective — a variance-tilted $(1-p)^\alpha u_N$ wins anytime performance while model-predictive replanning wins the final checkpoint, each losing the other’s metric (8–10/10 paired seeds each way; App. C). “Best curriculum utility” is ill-posed until that objective is fixed; we report both currencies throughout. The claim that carries the section is the pair of exact zeros at $p \in \{0, 1\}$, which every candidate utility shares — though the zeros alone do not suffice: a uniform-over-band control (sample the in-band indicator, discard the shape) forfeits most of the allocation gain on chains (0.681 vs. 0.743 AUC, 0/10 seeds; artifact `results_bridge.json`). u_N ’s standing is therefore: the shape matters, the best shape is objective- and testbed-dependent, and u_N is the one shape that ties the exact first-order objective as a predictor, requires no tuning, and transfers to the maze within noise.

Remark 2 (The curriculum changes the training distribution). Sampling prompts from an adaptive q_t instead of the data distribution ρ optimizes a reweighted objective $\mathbb{E}_{x \sim q_t}[J(x)]$: each group’s gradient is an unbiased MaxRL gradient *for its own prompt*, but the mixture is tilted toward the frontier, without importance correction back to ρ . This is the standard bargain every curriculum strikes (uniform sampling makes the same choice for $q_t = \rho$), and we treat it as an empirical question, not a free lunch: all evaluations in §6 are on fixed held-out pools under ρ , so any mismatch cost is charged to the method. The safety results are precisely about when this tilt helps or hurts the ρ -measured outcome.

One identity, three literatures: learnability curricula [11, 31] are the $N=2$ slice of u_N ; DAPO-style dynamic sampling and GRESO are its avoidance shadow (cull where $u \approx 0$); HER-style relabeling [17] is its creation complement (manufacture $K > 0$ where $u = 0$ identically; §5). DisCO [2] and SC-SDPO [3] derive analogous per-question weights for the GRPO family to diagnose difficulty bias; our identity generalizes that line to success-conditioned estimators and — the step neither takes — uses it to predict how *external* curricula and recyclers interact with each estimator, in coverage terms.

The three regimes, as a map of the paper (Fig. 3). The **teacher** (§4) allocates compute within the learnable region — and saturates: a true-pass-rate oracle allocator ties the cheap posterior in end performance (§6.1), so better pass-rate information is not where remaining gains live. **Recycling** (§5) creates signal where sampling the requested task yields none — in our control battery, the one intervention that still adds on top of the oracle. The **estimator** underneath shapes whether either is safe, and its mastered tail is where both failure modes live (§6). One line: *the teacher allocates, hindsight creates, the estimator decides whether either is safe* (Fig. 2).

4 FrontierMax: the schedule

FrontierMax is a standard group-rollout RLVR loop with three modified lines (Alg. 1). A decayed Beta posterior tracks each prompt’s pass rate from observed group outcomes only — never from relabeled successes, which inflate it ($\hat{p}$ 0.81 vs. eval 0.47). Thompson sampling draws $\tilde{p}$ and samples prompts $\propto u(\tilde{p})^\gamma$ with a uniform floor. The knobs (decay, floor, concentration, and relabel admission) remain empirical and task-dependent (Table 4); what is *derived* is the difficulty band itself — location, width, and N -scaling — which is exactly where competing curricula spend their hyperparameters.

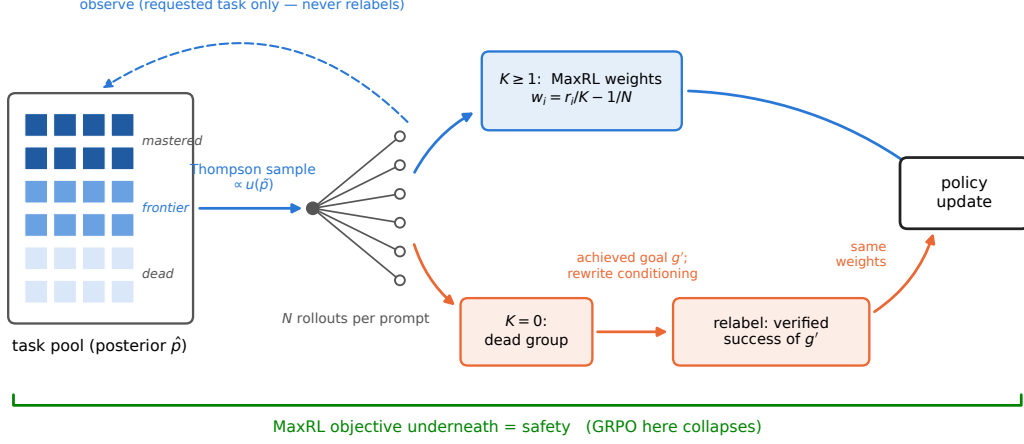


Figure 2: **One FrontierMax step.** A Thompson-sampled posterior allocates rollouts by the derived utility; live groups train with unmodified MaxRL weights; dead groups are relabeled to the sub-goal they actually achieved (conditioning rewritten) and train as verified successes. The posterior observes requested-task outcomes only.

Algorithm 1 FrontierMax: one step. Three lines differ from a standard group-rollout loop (marked $\triangleright$).

- 1: $\triangleright$ **sample** prompts $x_1..x_B \propto u_N(\tilde{p}_x)^\gamma$ mixed with a uniform floor, $\tilde{p}_x \sim \text{Beta}(\alpha_x, \beta_x)$ (else: uniform)
- 2: roll out N completions per prompt; verify; $K_x \leftarrow$ successes
- 3: update posterior (α_x, β_x) from (K_x, N) *for the requested task only*
- 4: live groups ($1 \leq K_x$): MaxRL advantages $w_i = r_i/K - 1/N$, unchanged
- 5: $\triangleright$ **dead groups** ($K_x=0$): relabel each parseable failure to *its own* achieved sub-goal g' (verified; conditioning rewritten); unparseable rollouts stay failures, so the group retains contrast ($K < N$)
- 6: $\triangleright$ admit the relabel only if $\hat{f}_{g'} \leq \text{gate_max_p}$, where $\hat{f}_{g'}$ is a decayed frequency of g' in the recycler's recent relabel stream (recorded whether or not admitted) (the gate: refuse recently seen destinations; never-seen destinations start at the Beta(1,1) prior $\hat{f} = .5$, admitted at the default threshold)
- 7: one policy-gradient step on live + admitted relabeled groups

5 Failure recycling

Remark 3 (Hindsight update, characterized). Relabeling a dead group to the sub-goal g' its trajectories actually achieved, rewriting the goal conditioning, and applying the same success-conditioned weights yields a *verified weighted-likelihood update for the achieved task*: every relabeled example is a true success of g' under the environment's verifier. It is generally an *adaptively selected, off-policy auxiliary update*, not an unbiased gradient of g' 's on-policy objective: g' is selected from the same group being scored, conditioning on the requested task's failure correlates the rollouts, the conditioning rewrite changes the law under which the log-probability is evaluated, no importance ratio corrects for it, and the finite- N weights target J_{N-1} (Lemma 1) rather than exact ML. It equals the fresh-rollout J_{N-1} gradient for g' only under a distribution-matching condition — the selected, rewritten group must have the same joint law as an i.i.d. fresh rollout group for g' . This specializes known relabeling-bias structure — the induced-distribution view of Eysenbach et al. [22] and the conditional-law gap in Ghosh et al. [23] — to success-conditioned RLVR weights; we state it as a remark, not a result, and note that the sharpening phenomenon of §6.8 runs through the *marginal* shift those works also identify, not this conditional matching. One further gap separates this remark from the deployed LLM loop: the remark is written for a group with *one* destination g' , but Algorithm 1 relabels each parseable failure to *its own* achieved goal, so one group can mix rewritten tasks — and once destinations differ within a group, the shared success count K couples unrelated tasks and the update is not a J_{N-1} gradient for *any* single achieved task. We therefore read the deployed per-row variant as a *verified weighted-SFT update*, treat

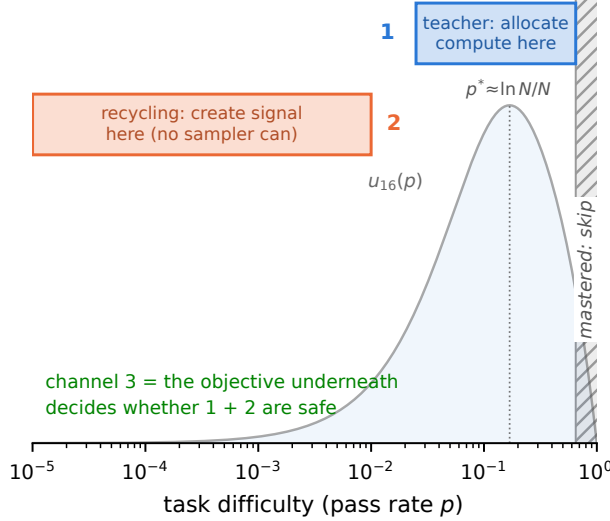


Figure 3: **The three regimes as a map of the method.** The teacher allocates compute within the learnable region; recycling creates signal where sampling the requested task yields none; the gate refuses relabels that land in the mastered tail; the estimator underneath decides whether any of it is safe.

its results as evidence about that objective, and measure the coupling’s cost where all variants are exactly implementable: per-row relabels as their own $K=1$ groups lead (AUC .952), one-destination-per-group is next (.881), and the shared- K coupled form forfeits most of recycling’s value (.749 vs. no-recycling .705; both orderings 10/10 paired seeds; artifact `results_row_vs_group.json`). Full object-level characterization, the CPU-reference contrast, and the LLM-side ablation flag: App. A.

Interpretation. “Exactness” is a property to measure per environment, not a blanket guarantee. We measured the two gaps the remark names where they are measurable (exact chains, 5 seeds; artifact `results_proposal_shift.json`): the applied relabel update’s cosine against the exact fresh-destination gradient is 0.93; the selection bias on the destination’s success rate is small here (+.003); and cross-fitting the destination choice removes the adaptive reuse at a real cost (alignment $\rightarrow$ 0.82, AUC .883 $\rightarrow$.829). The off-policy caveat is real; its sign is environment-specific.

Two contracts keep practice on the proof’s side: **(1) exactness** — a relabeled success must be a true success of the relabeled task under the environment’s own verifier (never an LLM judge); **(2) conditioning** — goal-embedding trajectories must have their conditioning rewritten to the achieved goal. Skipping (2) turns the exact gradient into noise: on our gridworld, hindsight without the rewrite scores below teacher-only ($0.600 < 0.658$); with it, it leads (0.703). Among the interventions studied here, recycling is the one that *creates* signal for an all-fail requested task (adding the achieved sub-goals to the pool by hand would too — what recycling buys is discovering them, §6.2), and we decompose its headline gain rather than let it inflate: of the +0.22 AUC on fixed task pools, a placebo replaying *live* gradients on dead-group slots captures 83% — most of the benefit is extra gradient dose on otherwise-wasted slots — while the relabel *direction* carries +0.037 beyond the strongest placebo, is the tightest arm in the battery, and is exactness-dependent (random-target relabels lose to dose-matched replay 5/5). Both numbers ship together — and the same replay control, transplanted to LLM scale as a pre-registered dose arm, makes the sign starker: there it exceeds recycling on *both* meters, 3/3 seeds (§6.8). The value is regime-dependent in a way we can state precisely: +0.22 AUC on fixed task pools, +0.01 on one-shot task streams — the gain is proportional to how much a relabeled skill can compound, and task-set fixedness is the single most important regime variable we found.

6 Experiments: an escalating ladder

Each rung asks one question of the three-regime structure, at increasing scale and decreasing experimental control; Table 1 is the map. Statistical contract, stated once and used throughout: the independent unit

Table 1: The ladder at a glance: every rung asks one question of the three-regime structure; scale rises, control falls.

rung	question	setting	seeds	outcome
6.1	does allocation saturate?	skill chain (exact grads)	5	yes: oracle ties posterior
6.2	anything to sample toward?	frontier-heavy pool	5	no: only creation scores
6.3a	does the curriculum pay?	maze cohort, 1.26M	3	throughput + recycling signal
6.3b	estimator effect, real grads?	maze factorial	6 blocks	endpoint failed; fresh AUC 6/6/sampler
6.4	where/when does coverage go?	maze, level-resolved	22 runs	descriptive early erosion
6.5*	when does allocation not pay?	IsaacLab Anymal-C	1	learnable-everywhere: null
6.6*	beyond task spread?	MountainCar/CartPole	3	spread=existence; tea.=speed
6.7	LLM scale?	GSM8K 2×2, 360M	1–2	delivery gate failed; inconclusive
6.8	what does recycling cost?	Countdown v2	3	mean up, bootstrap proxy down
6.9	does the gate recover it?	Countdown v2	3	corrected-code gate failed

*boundary rungs, compressed to one paragraph in the main text; full protocols and figures in App. C.

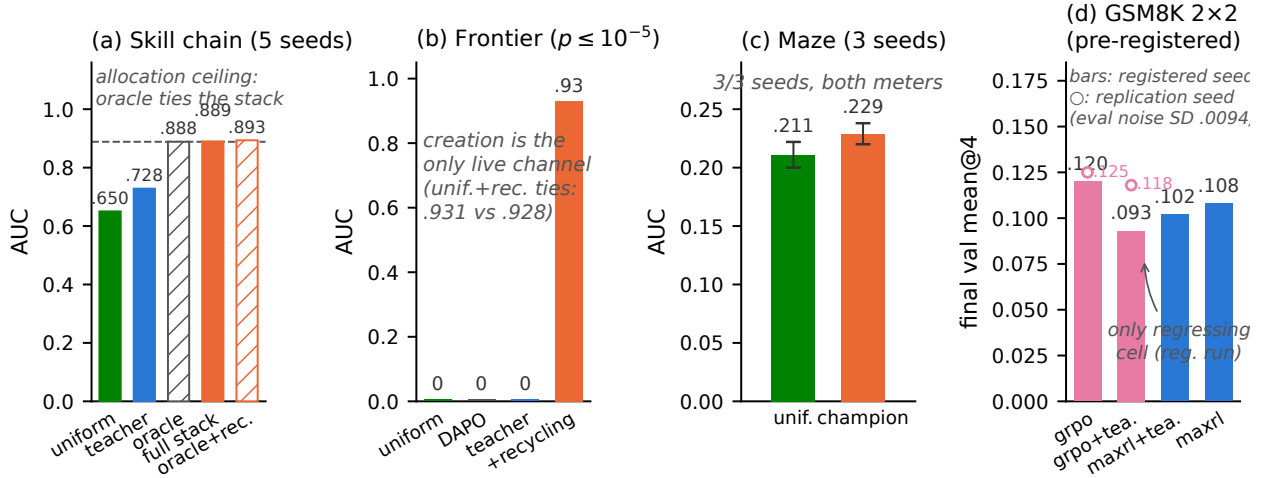


Figure 4: **The ladder.** (a) Exact-gradient chains: a no-floor, γ -matched true-pass-rate oracle matches the full stack within the observed five-seed variation — allocation saturates — and recycling still adds on top of the oracle. (b) Frontier-heavy regime: allocation scores exactly zero; recycling solves it, and does so equally under uniform sampling (.931 vs. .928) — creation, not allocation, is the live channel. (c) Maze at matched wall-clock: 3/3 seeds on both meters. (d) GSM8K 2×2: bars are the registered run, open markers the replication seed (GRPO cells; eval-noise SD .0094 quoted in-panel): GRPO+teacher is the only regressing cell on the registered run (pre-registered P-G2; the replication seed did not reproduce the shape — §6.7); the teacher’s MaxRL effect is a null at this budget (P-G1).

for any training-method claim is an independently trained seed block; runs sharing a seed, warmstart, or checkpoint are paired or correlated observations, never additional replicates; levels, k values, metrics, and checkpoints from one trained model are repeated measurements of it; repeated-evaluation spread measures evaluation noise, not training-seed uncertainty, and every $\pm$ names its source at point of use ($\pm$ with no qualifier is standard deviation across training seeds). Except where explicitly labeled otherwise, “coverage” is standard pass@ k with exactly k samples drawn per prompt and averaged over the evaluation pool. The historical Countdown quantity is instead VERL’s with-replacement bootstrap best@ k proxy and is never relabeled as pass@ k . Mean@ k is the mean success fraction of the same k samples; AUC is the mean of the evaluation curve over training at matched wall-clock or episode budget, and final is the last checkpoint. Where run pools are heterogeneous we say so and report signs descriptively rather than attaching p -values (§6.3–6.4). Pre-registered predictions were committed before the deciding runs; negative results appear in place, one sentence each, with full postmortems in the repository’s audit trail.

6.1 Does allocation saturate? (exact-gradient skill chains; CPU, 5 seeds). Uniform $0.650 \rightarrow$ teacher $0.728 \rightarrow$ full stack 0.890 ± 0.002 AUC (Fig. 4a). An eight-arm control battery (App. C) orders into a *gradient-information ladder*, each rung a 5/5 paired-seed win: zero-information dose ($0.798\text{--}0.800$) $<$ replayed correct direction (0.832) $<$ true relabels (0.869) $<$ perfect allocation (oracle $0.8885 \approx$ full stack 0.8895) $<$ allocation+creation (0.8935). Two decompositions matter downstream. *Allocation saturates at the top of the ladder, not at the teacher*: a no-floor, γ -matched true-pass-rate oracle (0.8885 ± 0.0014) sits within the five-seed variation of the *full stack* (0.8895 ± 0.0023) — similar within observed noise, not tested for equivalence. The posterior-only teacher scores 0.728 , far below the oracle, so perfect difficulty information is worth a great deal over the posterior alone; the accurate reading is that *recycling compensates for the posterior’s allocation error* — the combination reaches what perfect information reaches, by a different route. *Creation decomposes*: a replay placebo captures 83% of recycling’s $+0.22$; the relabel *direction* carries $+0.037$ beyond it and is the most reliable arm in the battery, while random-target relabels lose to dose-matched replay 5/5. The utility form matters through its N : the N -blind learnability slice $p(1-p)$ forfeits the entire allocation gain (-0.007 vs. uniform) where u_N carries $+0.078$ (5/5). All arms converge within $.027$ of one another — on fixed pools these interventions buy *time-to-ceiling*, not the ceiling.

Takeaway. A monotone information ladder: dose $<$ direction $<$ allocation-ceiling $<$ +creation. Allocation saturates; creation adds on top; every intervention buys speed, not asymptote.

6.2 Can any sampler act when nothing is learnable? (frontier-heavy regime, max pool pass rate 10^{-5} ; 5 seeds, artifact committed). Uniform, DAPO, and the plain teacher score **exactly 0.00**; adding recycling reaches **0.98 final (0.93 AUC)** — relabeling invents the curriculum below the pool, ignites within ~ 400 groups, then goes silent. The attribution control pins down which channel did it: *uniform*+recycling scores the same (0.931 ± 0.002 vs. teacher+recycling 0.928 ± 0.003 AUC) — in this regime the entire effect is signal creation; allocation contributes nothing because there is nothing learnable to allocate toward. Equivalently: a designer who hand-added the achieved sub-goals to the pool would also escape the zero — what recycling buys is *discovering* that sub-pool from the policy’s own failures, with no task engineering. The one loss-level alternative — the full control-variate estimator, which is *not* silent at $K=0$ — also scores 0.000 in every seed on this pool despite spending 100% of its update norm on all-fail groups: the all-fail term pushes every sampled action down symmetrically and cannot ignite a pool this deep (5 seeds, matched budget; artifact `results_fullcv_baseline.json`; detail in App. C).

Takeaway. Where nothing is learnable at budget, sampling policy is irrelevant — uniform and teacher tie at zero without recycling and tie at 0.93 with it. Even the estimator family’s nonzero all-fail update (full control variate) stays at zero here: creation is the only channel we tested that ignites a dead pool, and its value is discovery, not allocation.

6.3a Does the curriculum pay on the maze? (transformer at matched wall-clock; 1.26M params, 13 levels, 3 seeds). Champion (teacher + dense recycling) 0.252 ± 0.005 final / 0.229 ± 0.009 AUC vs. uniform 0.230 ± 0.015 / 0.211 ± 0.011 ; paired deltas positive in 3/3 seeds on both meters (six correlated contrasts, not six seeds); 22–35% more optimization steps per GPU-hour. Attribution: matched *wall-clock* lets curriculum arms take $1.2\text{--}1.4\times$ more optimization steps (dead groups skip the backward pass), and step-matched truncation shows the extra steps are most of the pure teacher’s effect (step-matched teacher-only $\Delta\text{AUC} +0.006$, within seed noise) while teacher+recycling survives step-matching, positive in all three paired seed blocks³ — on the maze the teacher’s measured value is *throughput*, recycling’s is signal. Utility-form controls (3-seed reruns, artifact `un_form_verdicts.json`): the exact u_N measures within noise of the legacy deployed form on the maze itself, and a CPU-winning $(1-p)$ -tilted refinement loses 0/3 here — within-band utility choices are testbed-sensitive, which is why the theory claims only the band’s boundaries.

6.3b Does the estimator effect survive real gradients? (the pre-registered factorial). An earlier nine-run cohort showed perfect sign separation — every MaxRL-labeled run grew pass@8, every GRPO-labeled run lost it — but the pool mixed sampler configurations and hindsight doses and reused warmstarts,

³A paired t at $n=3$ gives $p \approx .04$, reported as descriptive given its normality sensitivity; an exact sign-flip test cannot go below .25 at three blocks.

so we pre-registered the clean test: a balanced $\{\text{MaxRL}, \text{GRPO}\} \times \{\text{uniform}, u_N\text{-teacher}\}$ factorial, six independent seed blocks, identical per-block warmstarts, matched generation-and-optimization budgets (250 fixed steps), one registered endpoint (paired Δ mean pass@8, final minus warmstart), and a committed falsification branch. **The endpoint failed: 3/6 paired blocks under uniform, 4/6 under the teacher, and only 4/12 MaxRL cells grew coverage at all — the cohort’s zero-exception divergence is retracted, not softened** (prereg + raw cells + verdict: curriculum_maxrl/maze_gpu_factorial/, vendored with execution-commit provenance). The cohort pattern had conflated recycling’s coverage contribution with the estimator effect: its MaxRL runs included hindsight arms, and the factorial isolates the estimator.

What survives, at its evidence level. *Confirmed, exact rung — at matched learning rate:* frozen realized schedules replayed under each estimator preserve MaxRL over GRPO in 10/10 seed $\times$ schedule pairs at the common deployed lr (artifact `results_schedule_matched.json`). A pre-registered lr sweep then tested whether this is an lr-calibration artifact (reviewer-posed; P-LR1 — falsified as registered, branch executed), and on this learnable-everywhere pool it partly is: a global 2–4 $\times$ lr raise lets GRPO match MaxRL’s $\Delta\text{cov@8}$ — by saturating pass@1, so its coverage–reliability premium collapses to .007–.013 where MaxRL holds .050 (artifact `results_lr_matched.json`). Per the committed branch we state the exact-rung ordering as an at-matched-lr fact and carry the normalization caveat as a limitation: on pools where everything is learnable, more learning of any kind buys coverage, and only the premium separates the estimators; the neural-scale claim (below) is the one tested at matched budgets *and* identical optimizer settings — and its premium version also holds there (exploratory, computed after the lr sweep identified the premium as the invariant currency: time-integrated premium delta positive in 12/12 independent seed blocks after averaging repeated sampler contrasts; artifact `block_reanalysis.json`). *Found exploratorily, then registered and confirmed on fresh randomness:* on time-integrated coverage (mean in-training pass@8 minus warmstart), MaxRL retains more coverage than GRPO in **6/6** wave-1 seed blocks after within-block sampler averaging (and 6/6 under each sampler separately) — a noise-integrating secondary we had not registered, so we pre-registered it as the primary of a confirmation wave on six fresh seed blocks (P-F2, $\geq 5/6$ under both samplers, committed falsification branch) and it **confirmed at 6/6 under each sampler** (exact two-sided sign $p=0.031$ per sampler; means $+0.015/+0.024$). The easy-band localization met P-F3’s registered pair-level bar (10/12 repeated sampler contrasts), but those observations share six seed blocks: the block-level read is four positive, one tie, one negative, with a 95% interval crossing zero, so localization is suggestive rather than established. Across both waves the covAUC ordering is positive in 12/12 independent seed blocks. The retired *endpoint* contrast, unregistered on the fresh blocks, landed 5/6 there — consistent with the power analysis (§9) and left retired; the claim this paper makes is the time-integrated one.

Three controls close alternative readings. *Scheduler mismatch:* GRPO driven by its *own* exact mass functional ($\frac{1}{N}\mathbb{E}\sqrt{K(N-K)}$, Thompson-sampled) does not close the gap on chains (5/5) and loses coverage like every other GRPO arm in the factorial (5/6 cells negative, mean -0.042 , equal to uniform-GRPO). Thus this tested sampler does not explain away the estimator-conditioned ordering. *Variance normalization:* on chains, removing GRPO’s SD normalization drops it to RLOO’s coverage profile exactly as the mass account predicts (.148 vs. RLOO .161; sample-SD .762) — but in the factorial the no-SD arm loses the easy band in 6/6 seeds, indistinguishable from sample-SD GRPO, so **at neural scale the easy-band liquidation is not explained by variance normalization alone:** the Fig. 1 mechanism holds per-task under exact gradients and does not transfer through shared parameters unmodified (committed revision, falsification branch P-G0c). *Interaction:* the teacher does not amplify GRPO’s coverage loss in the factorial (teacher–uniform $\Delta\text{cov@8}$ under GRPO: mean $+0.015$, mildly protective) — the GSM8K P-G2 regression pattern receives no corroboration from the maze at this budget. Inference currency (single checkpoint pair; convention and per-level detail in App. C): our checkpoint needs **11 $\times$** fewer samples at level 5, ties GRPO at levels 2–3 — the summed-coverage curves cross at $k\approx 4$, so choose the estimator by how many samples you will draw at inference.

6.4 Where and when does GRPO’s coverage go? (level-resolved; cohort data — read with §6.3b’s retraction). This section decomposes the *original heterogeneous cohort* by difficulty band; after

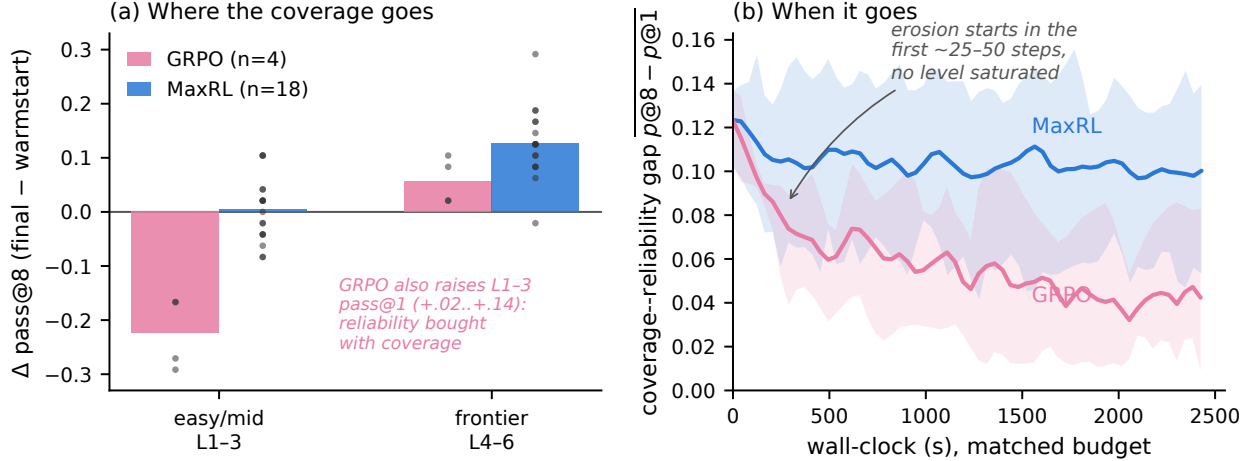


Figure 5: **Where and when GRPO’s coverage goes (maze, level-resolved, matched wall-clock).** (a) $\Delta \text{pass@8}$ (final – warmstart) by difficulty band: GRPO’s loss lives on the easy/mid band (all 4 runs; dots are runs) while its pass@1 there rises — sharpening; MaxRL holds the easy band and gains on the frontier band (18 runs; a heterogeneous run inventory — descriptive, no p -value). (b) Coverage–reliability gap $\text{pass@8} - \text{pass@1}$ vs. wall-clock (mean and min–max across runs): GRPO’s erosion begins within the first ~ 25 –50 steps, before any level saturates; MaxRL stabilizes after warmup.

the factorials, its value is the location and timing of GRPO’s loss (corroborated by the pair-level easy-band reads, with wave-2’s *registered* P-F3 meeting its 10/12 bar but remaining suggestive under block-level aggregation), not the retired zero-exception contrast (Fig. 5). *Where*: GRPO’s pass@8 loss is concentrated almost entirely on the easy/mid band L1–3 ($-.224$ mean over its four runs, every run negative) while its L1–3 pass@1 *rises* ($+.02$ to $+.14$, every run) — it funds reliability by liquidating easy-band coverage; MaxRL holds L1–3 flat ($+.005 \pm .058$ over 18 runs) and concentrates its gain on the frontier band L4–6 ($+.126 \pm .063$ vs. GRPO’s $+.057$). The 18-run MaxRL pool is a heterogeneous run inventory (teacher variants, hindsight doses, a wide model, repeated seeds), so we report the band asymmetry descriptively — every GRPO run negative on L1–3, the MaxRL spread centered at zero there — and attach no permutation p -value; the 22 runs are not exchangeable replicates of one estimator contrast. This is consistent with the mass curves in Fig. 1 — the $\sqrt{K(N-K)}$ weighting keeps spending updates on mastered prompts where u_N has already released them — but the step from mass to coverage is a stated *hypothesis*, not a derivation: mass is sign-blind, and “mastered-tail updates sharpen” requires an argument about the sign structure of near-saturated groups that we support with the location and timing evidence here rather than prove. Its committed falsifier (remove variance normalization and the tail signature should vanish) split across the rungs, as §6.3b reports: the mechanism account is per-task-exact only, and we bound the claim accordingly. *When*: the divergence is immediate, not saturation-triggered — GRPO incurs 59–73% of its total coverage–reliability gap loss ($\overline{\text{pass@8} - \text{pass@1}}$: $.123 \rightarrow .042$) within the first quarter of the budget, with the first easy-band pass@8 drop visible at steps 0–50 when L1 pass@1 is still only $.62$ – $.75$; MaxRL’s premium stabilizes after warmup ($.122 \rightarrow .101$). A dose-response note from the same logs (descriptive, same caveat as above): sparse relabeling (1/dead group) uniquely preserves the coverage–reliability gap ($\overline{\text{pass@8} - \text{pass@1}}$) relative to none and dense, while dense banks the same coverage into the best final pass@1 — dose sets *where* the coverage dividend is spent, not whether coverage is lost.

Takeaway. The easy band is where the estimators differ — GRPO liquidates it for reliability from the first updates, and both factorial waves keep that asymmetry (wave-2’s registered P-F3: 10/12 repeated sampler contrasts, but only four positive blocks, one tie, and one negative after within-block averaging) — but the mechanism attribution to variance normalization holds only at the exact rung. Coverage is the currency that sees all of it.

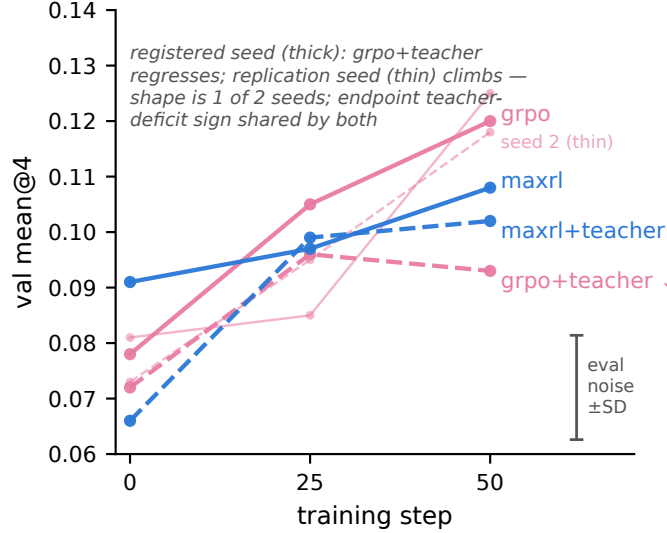


Figure 6: **GSM8K 2×2, val mean@4 over training.** Both GRPO seeds shown as paired trajectories (registered seed thick, replication thin); gray bar: repeated-eval noise floor of one fixed checkpoint ($\pm$ SD .0094), drawn once as a scale. On the registered run GRPO+teacher is the only cell that regresses in its second half; the replication seed climbs instead. §6.7 gives the statistics and the steering telemetry that separates the two seeds.

6.5–6.6 Two boundary rungs (compressed; full protocols, figures, and attribution in App. C).

An IsaacLab Anymal-C pilot (5 arms, one seed, independent team, pre-registered null) marks where allocation does *not* pay: on a terrain grid learnable everywhere at budget, the stock greedy curriculum beats the frontier teacher outright (.372 vs. .278) — with the maze step-matched analysis and Countdown, the regime pattern is **allocation pays where unlearnable-at-budget regions exist to avoid**. A Gymnasium control suite (binary task success, 3 seeds) marks the opposite boundary: training on the target task alone is a provable zero-update freeze (every group all-fail, so every group-relative estimator emits exactly zero), while the gated full stack reaches 1.000 ± 0.000 on both official tasks (Fig. 9, appendix); attribution shows task spread supplies gradient existence, the teacher speed and stability, and a per-bin-parameters ablation reproduces the known failure — **curricula operate through shared parameters, or not at all**.

6.7 Does it transfer to LLM scale? (GSM8K 2×2, pre-registered; SmolLM2-360M, $N=16$, one A10G). Predictions committed before any cell finished. **P-G2 landed on the registered run (Fig. 6):** GRPO+teacher was the only cell that regressed in its second half; the deciding statistic is the endpoint deficit against plain GRPO, 2–4× the measured same-model eval noise (mean@4 $-.027$; pass@4 $-.048$), while plain GRPO climbed monotonically (.078 $\rightarrow$.105 $\rightarrow$.120).⁴ **A replication seed does not reproduce the regression shape:** seed 2 of the same cell climbs monotonically (.073 $\rightarrow$.095 $\rightarrow$.118) under measurably weaker steering (near-uniform teacher on average; telemetry in App. C). The completed within-seed pair (grpo-uniform seed 2: .081 $\rightarrow$.085 $\rightarrow$.125; artifact `grpo_uniform_seed2.json`) sharpens the read: the endpoint teacher deficit under GRPO has the same *sign* in both seeds on both (coupled) meters, with magnitude tracking steering intensity — a dose effect, not a sign flip; the pre-registered second-half-regression *shape* remains 1 of 2 seeds, suggestive not established. MaxRL+teacher trains stably to its best value (.066 $\rightarrow$.102) but does not beat uniform’s .108 — P-G1 is a *null at this budget*, predicted in advance by posterior starvation: the teacher’s posterior learns this model’s real difficulty landscape from ~ 1 visit per prompt, but 3,200 draws over 7,473 prompts leave the estimated utility nearly flat, so Thompson sampling stays near-uniform and the delivered treatment was weak in every cell (run-mean dead-sampled

⁴The second-half dip itself (.096 $\rightarrow$.093) is within the $\pm .009$ noise floor and carries no weight on its own. One caveat governs every GSM8K number: the $\pm .009/\pm .017$ bars are repeated-eval noise of a fixed checkpoint — they bound evaluation variability, not training-seed variability — so deficit-to-noise ratios are descriptive reads, not hypothesis tests.

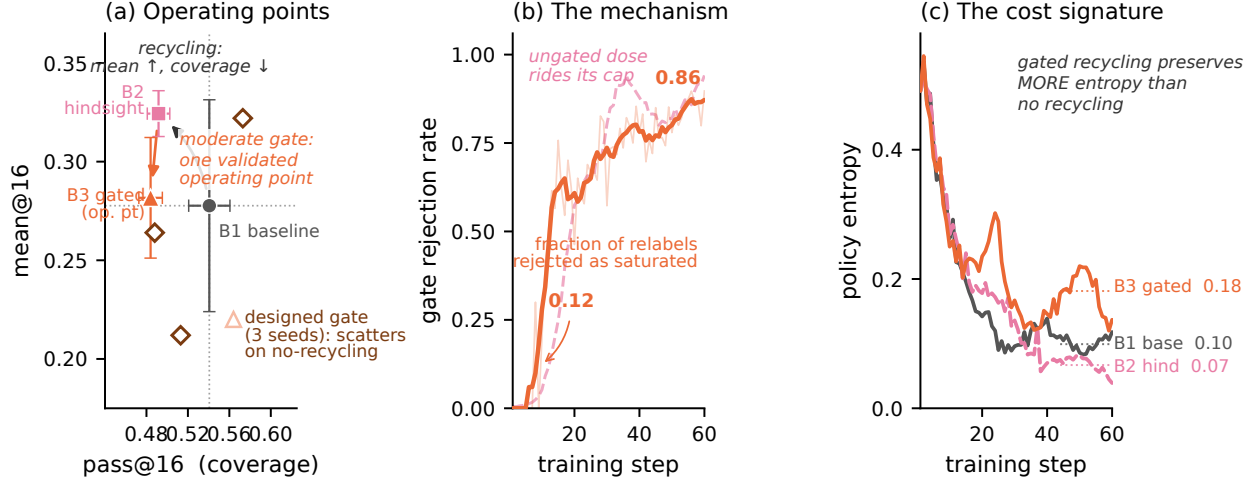


Figure 7: **Recycling-package sharpening and the unvalidated saturation gate (Countdown, tier 1).** (a) Reported three-seed aggregates: recycling (B2) raises mean@16 while the logged VERL bootstrap best@16 proxy falls; the moderate gate (B3) gives most of the mean back on this early-saturating tier but used faulty decay. The pre-registered designed-strength arm (diamonds; 3 seeds, corrected decay, rejection $\sim .94$) refuted the strength-dial reading: its seeds scatter around the no-recycling point rather than extending a frontier (P-R1, §6.9), so it is drawn as a scatter per the committed falsification branch; the superseded 1-seed point (faint) sits inside that spread. (b) The mechanism in-run: the gate’s rejection rate rises monotonically as the model’s reachable set saturates its relabel destinations, while the ungated dose rides its cap. (c) Gated recycling preserves *more* policy entropy than no recycling — admitted relabels are a spread-preserving gradient.

fractions indistinguishable; diagnostics in App. C). A pass@ k sweep on the registered checkpoints has the teacher’s GRPO deficit *widening monotonically with k* ($-.008$ at $k=1$ to $-.024$ at $k=16$) — curriculum damage lives in coverage, as the maze predicted, though each delta is within single-seed noise. One design confound is disclosed rather than dismissed: teacher cells sample with replacement and so revisit prompts more. A steering-strengthened follow-up committed an automatic treatment-delivery gate before its endpoint: minimum dead-sampled fraction $< .50$ and run mean $< .60$. It completed with minimum .413 but run mean .60148, failing the conjunctive gate by .00148; under the committed branch its endpoint is inconclusive by design and the dependent repetition/safety contrasts cannot establish the interaction. Small model; the registered outcomes remain orderings and signs on one run, not a replicated LLM effect.

Takeaway. Prompt-level posterior steering was not reliably delivered at this scale and budget; the LLM interaction is not established and the maze factorial does not corroborate it.

6.8 Historical Countdown development result (superseded). The original 2×2 exact-verifier experiment returned two null predictions because its pool had little headroom. Its single-run concentration pattern motivated the rebuilt three-seed study below, but it is not the applied claim. The metric originally labeled pass@16 is VERL bootstrap best@16 and must be read as a coverage proxy under the erratum above.

An externally specified higher-dose arm later applied two optimizer epochs to every live group with hindsight disabled. It reached mean@16 $.478 \pm .021$ and bootstrap best@16 $.635 \pm .046$. Recycling permits at most 12 auxiliary groups per 64 requested groups, whereas this replay arm doubles epochs on every live group. It is therefore a higher-dose alternative that improves both logged metrics, not a dose-matched control, relabel-specific effect, or causal bound. Complete relabel-arm seed records and per-task outcomes remain missing.

Takeaway. The original pool is a development null. The higher-dose replay arm does not identify the effect of relabel direction.

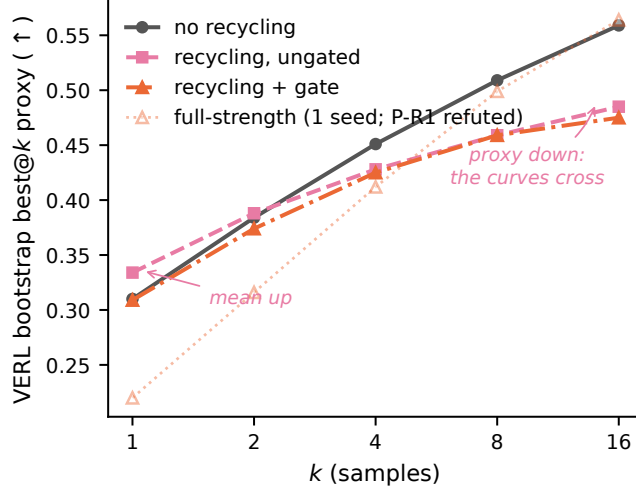


Figure 8: **Exploratory single-seed VERL bootstrap best@k proxy (Countdown tier 1, step-60 val, $n=128$ tasks)**. This with-replacement metric is not standard unbiased pass@ k . Ungated recycling sits above baseline at $k=1$ and below by $k=4$ in the logged proxy, with the deficit widening with k . The moderate gate tracks baseline at low k with a smaller high- k deficit; the full-strength curve (hollow; one seed, corrected decay) pays at $k=1$ and finishes above baseline at $k=16$, but the strength-dial reading this suggests was refuted in the 3-seed designed-strength sweep (P-R1, §6.9). The single-seed points and session provenance are stored in the declared figure input; they are exploratory and cannot be converted without the underlying task outcomes.

6.9 Does the gate recover it? (Countdown v2 pool, 3 seeds, pre-registered). *The gate, and what it is not.* The rule we implement is a saturation heuristic: a decayed frequency statistic $\hat{f}_{g'}$ over the recycler’s own recent relabel destinations, thresholded at `gate_max_p`. It is *not* an estimate of the destination task’s fresh-rollout pass rate $p_{\theta}(\text{success} \mid g')$ — the gate never rolls out g' and never observes a g' failure — and the threshold 0.5 is a tuned midpoint, not a derived zero.⁵ Where destination pass rates are exactly measurable (skill chains, 10 paired seeds; artifact `results_gate_variants.json`), the true- p gate preserves essentially all of ungated recycling’s value (AUC .879 vs. .881) while the frequency heuristic pays a toll (.798; 10/10); at decision time the heuristic’s statistic *anti-correlates* with true $p(g')$ ($-.27$ — it tracks the recycler’s own recent output), and the two rules agree only once everything saturates ($.66 \rightarrow .99$). The results below are therefore evidence about a recency-novelty filter that approximates saturation late in training. The upgrade is also practical, not just derived: an *estimated- p* gate fed by probe rollouts of proposed destinations recovers 98% of the oracle-vs-frequency gap at matched *total* rollouts with even one probe per step (10/10 seeds at every probe budget tested, insensitive to the budget over 1–16; pre-registered P-PB1, confirmed; artifact `results_gate_probe_budget.json`) — destination pass rates are cheap to estimate because the gate only needs them coarsely (≤ 0.5), so the LLM-loop version costs a negligible slice of generation budget.

The experiment. We rebuilt the pool with real structure (random operand permutation, parenthesization, exact division; a headroom gate excluded guesser-saturated tiers) and ran three arms $\times$ three seeds: no recycling (B1), ungated recycling (B2), gated recycling (B3). One setup note kept for its theory reading: recycling has a *syntactic prerequisite* — the pre-SFT base model produces zero verifier-valid failures, so there is nothing to recycle until SFT installs the output format (pool construction and baseline calibration in App. C). **Aggregate concentration pattern:** at tier 1, recycling raises mean@16 by $+.046$ ($.278 \pm .066 \rightarrow .324 \pm .014$) while the logged VERL bootstrap best@16 proxy falls by $-.049$ ($.541 \pm .024 \rightarrow .492 \pm .013$), reported as mean $\pm$ sample SD across three seeds. Complete B1/B2 seed records are not vendored (seed 3 is aggregate-only), so we do not claim paired seed signs. Figure 8 shows seed 1’s exploratory logged proxy over k ; per-task outcomes are unavailable and standard unbiased pass@ k cannot be reconstructed.

⁵ u_N ’s exact high- p zero is at $p=1$, which would admit everything; a gate on a genuine task-conditioned $\hat{p}_{g'}$ is the derived object.

Available aggregate dynamics suggest that the mean increase precedes the proxy decrease, but complete seed-level records are absent. We therefore do not make paired timing, per-seed direction, or early-stopping claims.

The gate did not validate. The favorable moderate-gate setting used faulty decay code and remains descriptive. A corrected, externally specified strong-gate sweep landed near the no-recycling aggregate and refuted the claimed monotone gate dial; it does not establish equivalence. Rejection and entropy telemetry are consistent with saturation filtering but do not identify the cause of the aggregate pattern. A successful, fixed-code mitigation remains open.

Takeaway. The retained result is mean-up with a bootstrap coverage proxy down. The gate and the causal concentration mechanism are not validated.

7 Related work

The coverage tension. RLVR itself trades coverage for precision: pass@1 rises while pass@ k at large k falls below the base model [4], with entropy collapse as the mechanistic signature [6] and overtraining on saturated problems as a locus [5]. Our contribution to this line is the *interaction*: whether external curricula and recyclers compound or cancel the estimator’s implicit weighting, and what that costs in coverage. Notably, recent work frames curricula as a potential cure for coverage loss [7]; our controlled comparison instead varies the estimator under a common sampler. The estimator–difficulty interaction itself is a heating 2026 theme — SC-SDPO [3] shows GRPO’s normalization implicitly encodes a difficulty weighting, and A-GRAE [30] redesigns the advantage around difficulty focus. Our experiments isolate the estimator-conditioned coverage ordering under identical sampling interventions, a comparison not isolated in the cited studies.

Curricula for RLVR. Difficulty-adaptive sampling is bucket- or scalar-level and heuristic-scored: DUMP (UCB over buckets), AdaRFT (scalar controller over precomputed labels, TMLR 2026), SEC (category bandit), threshold curricula. LILO (NeurIPS 2025) is the closest per-prompt method — rejection sampling by $\hat{p}(1 - \hat{p})$, which Prop. 2’s corollary identifies as the $N=2$ slice of our derived utility. Reinforce-Ada [13] — the adaptive sampler MaxRL itself points to — adapts the rollout count *within* a prompt until a signal criterion is met, holding the prompt distribution fixed; our teacher is the complementary move, reallocating *across* prompts at fixed group size, and the static water-filling reference (App. B) is the idealized form of both. DAPO’s dynamic sampling and GRESO cull dead prompts — avoidance without allocation or creation. PKPO [15] and Pass@ k training [16] modify the objective toward pass@ k but keep uniform sampling. **Goal selection in HER.** Selecting *which* goals to train on by difficulty or reachability is a decade-old idea in the relabeling literature: intermediate-difficulty goal generation [18], value-based hindsight goal selection (HGG; 19), curriculum-guided HER’s proximity–diversity admission of relabel goals [20], and Skew-Fit’s α as a concentration-vs-coverage dial [21]. Our gate belongs to this mechanism class; what is new is not admission-by-difficulty but its *motivation* — the same estimator functional u_N used for task sampling vanishes in the mastered tail. The gate statistic itself is a tuned heuristic (a decayed achieved-destination frequency with a tuned threshold, §6.8–6.9), and its corrected-code test did not validate a useful operating point. We therefore claim a shared motivation and report a failed intervention, not a switch-free derivation or a working mitigation.

Hindsight and self-training for LLMs. Exact-verifier relabeling has precedent outside RLVR loops — CodeIt [26] (ARC program synthesis, relabel + prioritized replay) and proof-tree relabeling [27] — and LLM-judged variants relabel agent trajectories (AgentHER, HSL). Concurrent 2026 work brings relabeling-adjacent mechanisms into RLVR-style training: LfH [25] (a VLM proposes one shared hindsight instruction per failed group for VLA post-training; the relabeler is a judge, their stated limitation) and ZipRL’s hindsight response replay [28] (densifying GRPO signal for multi-turn agents). We did not identify a cited study that reports a pass@ k -style coverage metric under relabeling. Our surviving Countdown evidence is a logged bootstrap coverage proxy rather than standard unbiased pass@ k , so it motivates that measurement without yet completing it. The nearest phenomenon report is in same-problem rejection finetuning: Strozzi [29] document a STaR-trained model winning pass@8 while the base model overtakes at pass@64 — concentration over reach — for SFT on a model’s own verified outputs. The reported recycling-package trade-off is a cross-task, selected-auxiliary-update analogue, and our gate is a tested but unvalidated admission heuristic; gating *vanilla* RLVR updates by saturation signals has independent precedent (5’s Bayesian boundary gating), not

applied to relabel destinations. The precise claims we retain are narrower: (i) the sampling score is derived from the deployed estimator’s expected coefficient mass; (ii) the same estimator convention is used for the theory and implementation; and (iii) the Countdown record exposes a mean-coverage-proxy trade-off that requires a prospective raw-outcome pass@ k test. The current evidence does not establish a relabel-specific causal mechanism or a successful gate.

8 Preregistered boundaries

Two campaigns registered after this draft’s experiment ladder bound the claim, and both are reported in full in the ICLR submission body (`paper/body_iclr.tex`).

The peak location is not what helps. In every rung above, the winning arm is also the harder-peaked arm, so “harder peaks beat $p(1-p)$ ” and “the deployed- N peak is correct” predict identically — and only the second would advance on the $p(1-p)$ literature. Holding the deployed estimator at $N = 16$ and varying *only* the sampling exponent, an over-shooting u_{64} arm is not beaten by u_{16} ($-.0113$, 95% CI $[-.0297, +.0057]$, exact $p = .235$, 10/20; and $-.0128$, $[-.0284, +.0024]$, $p = .131$, 7/20 on a second platform). A seven-point sweep $N_{\text{score}} \in \{2, \dots, 128\}$ has its argmax at u_{64} , not at the deployed u_{16} (Spearman $+.93$ between exponent and mean). The value of u_N over $p(1-p)$ lies in its tail, precisely where w_{N-1} differs from 1 — not in the derived peak. Branching an exact-gradient pool to measure true continuation utility U_H at $H = 8$ shows why: u_{16} ranks U_H far better than $p(1-p)$ under MaxRL ($+.106$, exact $p = .002$) but not under RLOO ($+.0001$), while U_H itself peaks near $p \approx .02$, below both p_{16}^* and p_{64}^* . Activity ranks utility well and *locates* it poorly.

Activity is not a standalone replay signal. Dropping the posterior activity priority into `minimax`’s robust PLR in place of MaxMC, with the shipped maze configurations otherwise verbatim, is negative: upstream scores .629 and the no-curriculum control .539, while every activity-priority variant falls below the control (.500 at $N=8$, 0/5 seeds). The cause is bandwidth, not shape — MaxMC reads critic disagreement at every timestep, whereas at the shipped `n_eval=1` a success-rate score receives one terminal Bernoulli per level visit. Gating MaxMC by activity rather than replacing it recovers most of the gap (.590; $-.039$ to upstream, 1/5) without clearing it. The AMaze student is PPO with GAE and never forms binary rollout groups, so this bounds activity as a replay priority, not the estimator-derived mechanism itself.

9 Limitations and negative results

Scale, and what the retraction taught about design. The factorial retraction is reported in full in §6.3b; what belongs here is its power lesson. Computed from all 36 runs’ successive-eval differences, per-eval noise is $\sigma \approx .017$, giving the registered paired single-eval endpoint a contrast sd of .024 and power only 0.4–0.8 at the effect size the data itself displays ($+.025/+0.031$ paired means). The time-integrated version of the same contrast cuts that sd by $\sqrt{10}$ (power ≈ 1 at $+.02$), which is why it became the registered primary of the confirmation wave on six fresh seed blocks — where it confirmed (P-F2, 6/6 per sampler; §6.3b). The LLM-scale teacher×estimator interaction was 1-of-2 seeds and tracked measured steering intensity: the replication seed of the GRPO+teacher cell did not reproduce the registered regression, and its telemetry shows the treatment was barely delivered (§6.7); the maze factorial’s interaction read (teacher mildly protective under GRPO) does not corroborate it. The GSM8K teacher is also coverage-neutral-to-negative under *both* estimators at this budget, so the contrast there concerns the sign of the mean-accuracy effect, not a coverage rescue. A strengthened follow-up then failed its preregistered run-mean delivery gate; its endpoint is inconclusive, so we make no LLM interaction claim.

What the prospective Countdown control can identify. The frozen E2c design repairs the historical result’s most important measurement defects: it retains binary task outcomes, recomputes standard pass@16, pairs evaluation seeds and task manifests, and gates fixed-slot and response-token dose before exposing endpoints. It is still not a pure relabel-direction experiment. B2 uses current-policy failures rewritten to

achieved goals, whereas E2c replays informative groups from a frozen-SFT reservoir. Source age, behavior distribution, task selection, and on/off-policy character therefore differ even when delivered group and token counts match. A valid B2–E2c contrast identifies the achieved-target relabel *package* relative to the frozen informative-replay package at matched delivered dose. With three training seeds, the resulting treatment contrast remains descriptive; tasks and rollouts improve endpoint precision, not treatment replication.

Where the teacher cannot work as specified. Two structural limits at production scale, stated as design constraints rather than future work. *Posterior starvation is the generic regime*: a per-prompt Beta posterior needs multiple visits, but realistic RLVR pools see ~ 1 epoch ($\leq 2\text{--}3$ visits per prompt), so FrontierMax as specified is a small-pool instrument; deploying it on 10^5 -prompt pools requires coarser state — bucket-level posteriors, a learned difficulty predictor, or warm starts — and our external annotations transferred poorly ($\rho = -0.17$), so cold-start from off-model labels is not the answer. *The throughput mechanism does not transfer*: at matched wall-clock the maze teacher’s measured value is extra optimization steps from skipping dead groups’ backward passes, but in generation-dominated LLM pipelines (up to 83% of step time in our generation-dominated sessions; App. E) a dead group has paid its full cost before it is known dead — the relevant LLM-scale comparisons are generation-budget-matched (resample-to-fill, prompt pre-filtering), and the teacher’s value there rests on avoiding the generation itself, which requires the posterior it starves for. These two constraints compound, and we state them together because they bound the same deployment.

Scope of the surrogate. Coefficient mass is not gradient norm, SNR, or expected improvement (Remark 1) — it omits score-vector geometry, sequence length, token masking, optimizer state, and clipping — and §6.5 exhibits a regime where reallocating it buys nothing (learnable-everywhere pools). Recycled-gradient exactness is a property of the environment’s relabel map. The fixed-pool and one-shot results ($+0.22$ vs. $+0.01$ AUC) show strong regime dependence; they do not define an interpolation law for noisier verifiers.

The coverage main effect is pool-conditional. A pre-registered fifth suite (Jugs water-measuring puzzles; preregistration, per-cell raw endpoints, verdicts, and postmortem vendored under `jugs_llm/results/` with execution-commit provenance) returned all-null: on a pool whose learnable band is a single stratum of 1–2-move template solutions, *plain MaxRL* also collapses — entropy $1.36 \rightarrow 0.07\text{--}0.21$ in 60 steps, frontier-tier pass@16 $0.15 \rightarrow 0$, every arm converging to the same deterministic policy — and recycling accelerates the collapse rather than rescuing it. When N rollouts of a near-trivial plan are near-identical, the i.i.d.-rollout diversity the group contrast requires is absent, and success-conditioned weights liquidate frontier coverage through shared parameters just as GRPO’s do elsewhere. The estimator main effect of §6.3–6.4 is therefore a claim about pools with a graded band at the deployed N , not an unconditional safety guarantee; per-task rollout-set diversity at initialization is the design gate we now check first. The same suite also exposed a granularity bug in the gate (keying destinations by value collides tasks when achieved values are few and shared — ≤ 20 integers in Jugs vs. hundreds in Countdown); the corrected gate keys on the relabeled task.

What did not transfer. A $4\times$ duration run refuted our own duration hypothesis (the deep-frontier stall is a per-step-competence ceiling; depth needs capacity, not schedule); γ -concentration did not transfer from chains to the maze, as the compounding model predicted; adaptive truncation order underperformed fixed T ; and feeding relabeled successes to the teacher’s posterior inflates it, so the posterior sees requested-task evidence only.

Corrections along the way. Two early claims were retracted on internal review — a teacher-steering read traced to an epoch-frozen sampler (the fixed run confirmed the effect), and a “beats the oracle” comparison traced to a floor handicap in the oracle arm (§6.1 reports the corrected match-within-noise). Full postmortems live in the repository’s audit trail.

10 Conclusion

For the zero-stabilizer practical MaxRL estimator under i.i.d. binary rewards, the expected absolute coefficient mass of a prompt is exactly $2(\text{pass}@N - \text{pass}@1)$, and dropping all-fail groups targets truncation order $N-1$. This identity provides a rollout-aware activity diagnostic and a principled curriculum hypothesis. It does not make coefficient mass a universal learning objective: score geometry, parameter sharing, optimizer calibration, and task transfer remain outside the scalar calculation.

The empirical record supports that narrower interpretation. An externally recorded fresh maze wave finds a positive time-integrated MaxRL–GRPO coverage contrast in 6/6 blocks under each sampler, while the stronger endpoint claim and easy-band mechanism do not survive their cleaner tests. The Countdown aggregate raises mean accuracy while lowering a logged bootstrap coverage proxy, but missing task outcomes and failed matched controls leave its causal mechanism unresolved. The corrected saturation gate does not validate, GSM8K misses its treatment delivery criterion, and Jugs supplies a preregistered domain boundary.

The resulting practical recommendation is simple: evaluate a curriculum or recycler together with the estimator it modifies, and report coverage beside mean accuracy. The remaining decisive experiment is prospective rather than interpretive: retain raw task outcomes, verify matched treatment delivery, and compute standard $\text{pass}@k$ before making a recycling-specific causal claim.

References

- [1] Tajwar, F., Zeng, G., et al. Maximum Likelihood Reinforcement Learning. *ICML*, 2026.
- [2] Li, G., Lin, M., Galanti, T., Tu, Z., Yang, T. DisCO: reinforcing large reasoning models with discriminative constrained optimization. *NeurIPS*, 2025.
- [3] Liu, Z., Cao, Y., Chen, J., Honavar, V. G. Restoring the sweet spot: pass-rate weighted self-distillation for LLM reasoning. arXiv:2605.27765, 2026.
- [4] Yue, Y., et al. Does RL really incentivize reasoning capacity in LLMs beyond the base model? *NeurIPS* (oral), 2025.
- [5] Yuan, S., et al. Understanding diversity collapse in RLVR via the lens of overtraining. arXiv:2606.15455, 2026.
- [6] Cui, G., et al. The entropy mechanism of RL for reasoning language models. arXiv:2505.22617, 2025.
- [7] Cai, W., et al. Curriculum RL can incentivize reasoning capacity beyond the base model. arXiv:2606.22317, 2026.
- [8] Wang, Z., Cui, G., Li, Y.-J., Wan, K., Zhao, W. DUMP: automated distribution-level curriculum learning for RL-based LLM post-training. arXiv:2504.09710, 2025.
- [9] Shi, T., Wu, Y., Song, L., Zhou, T., Zhao, J. Efficient reinforcement finetuning via adaptive curriculum learning. *TMLR*, 2026. arXiv:2504.05520.
- [10] Chen, X., et al. Self-evolving curriculum for LLM reasoning. arXiv:2505.14970, 2025.
- [11] Foster, T., Sims, A., Forkel, J., Foerster, J. Learning to reason at the frontier of learnability (LILO). *NeurIPS*, 2025. arXiv:2504.03380.
- [12] Yu, Q., et al. DAPO: an open-source LLM reinforcement learning system at scale. *NeurIPS*, 2025. arXiv:2503.14476.
- [13] Xiong, W., et al. Reinforce-Ada: an adaptive sampling framework under non-linear RL objectives. arXiv:2510.04996, 2025.
- [14] Zheng, H., et al. Act only when it pays: efficient reinforcement learning for LLM reasoning via selective rollouts (GRESO). *NeurIPS*, 2025.
- [15] Walder, C., Karkhanis, D. Pass@ k policy optimization: solving harder reinforcement learning problems. arXiv:2505.15201, 2025.
- [16] Chen, Z., et al. Pass@ k training for adaptively balancing exploration and exploitation of large reasoning models. arXiv:2508.10751, 2025.

- [17] Andrychowicz, M., et al. Hindsight experience replay. *NeurIPS*, 2017.
- [18] Florensa, C., Held, D., Geng, X., Abbeel, P. Automatic goal generation for reinforcement learning agents. *ICML*, 2018.
- [19] Ren, Z., Dong, K., Zhou, Y., Liu, Q., Peng, J. Exploration via hindsight goal generation. *NeurIPS*, 2019.
- [20] Fang, M., Zhou, T., Du, Y., Han, L., Zhang, Z. Curriculum-guided hindsight experience replay. *NeurIPS*, 2019.
- [21] Pong, V. H., Dalal, M., Lin, S., Nair, A., Bahl, S., Levine, S. Skew-Fit: state-covering self-supervised reinforcement learning. *ICML*, 2020.
- [22] Eysenbach, B., Geng, X., Levine, S., Salakhutdinov, R. Rewriting history with inverse RL: hindsight inference for policy improvement. *NeurIPS*, 2020.
- [23] Ghosh, D., et al. Learning to reach goals via iterated supervised learning. *ICLR*, 2021.
- [24] Ferbach, D., et al. Self-consuming generative models with curated data provably optimize human preferences. *NeurIPS*, 2024.
- [25] Xu, I., Jiang, S., et al. Learning more from less: reinforcement learning from hindsight. arXiv:2607.09042, 2026.
- [26] Butt, N., et al. CodeIt: self-improving language models with prioritized hindsight replay. *ICML*, 2024.
- [27] Poesia, G., et al. Learning formal mathematics from intrinsic motivation. *NeurIPS*, 2024.
- [28] Hu, Z., Wang, L., Wang, X., et al. ZipRL: adaptive multi-turn context compression with hindsight response replay. arXiv:2605.28069, 2026.
- [29] Strozzi, I. L. Teacher-free self-training amplifies but does not compound: a pass@ K crossover on a free-verifier domain. arXiv:2606.07856, 2026.
- [30] Yu, Z., Chen, Z., Liu, M., et al. Unveiling implicit advantage symmetry: why GRPO struggles with exploration and difficulty adaptation. arXiv:2602.05548, 2026.
- [31] Rutherford, A., et al. No regrets: investigating and improving regret approximations for curriculum discovery. *NeurIPS*, 2024.

A Mixed-target relabel groups: full characterization

The deployed per-row LLM variant trains each relabeled row as a true success of its own rewritten prompt at weight $1/K - 1/N$, with two consequences we accept and disclose: the $-1/N$ failure weight acts as a conditional push-down on unparseable rollouts rather than a zero-mean baseline, and success weights couple across unrelated destinations. The CPU reference implementation instead trains each relabeled trajectory as its own $K=1$ group at weight $1 - 1/N$, which *is* the single-destination object of Remark 3; the per-row LLM variant was pre-registered as a deviation because Countdown groups rarely share achieved values. The single-destination MaxRL form — one destination per dead group, all N rows rewritten and reverified against it — is the design the remark licenses. Whether the deployed loop pays the chain-rung coupling penalty is a question that rung cannot answer (verl’s normalization and failure conditioning differ); the artifact ships a single-destination ablation flag (`one_target_per_group`, CPU-tested) and its pre-registered LLM run for exactly that test.

B The static allocation reference

Proposition 4 (Allocation, static). *Consider the static problem $\max_{\{N_i\}} \sum_i u_{N_i}(p_i)$ subject to $\sum_i N_i = B$ and integer $N_i \geq 1$ (every prompt receives a mandatory initial rollout; feasibility requires $B \geq$ the number of prompts), with pass rates p_i fixed and known and one group per prompt. Because the marginal gain $u_{N+1}(p) - u_N(p) = p(1-p)^N$ is decreasing in N , greedy water-filling of the $B - \#$ prompts remaining rollouts on this marginal — the probability that the next rollout is a group’s first success — is optimal. The restriction*

to $N_i \geq 1$ is not cosmetic: the formula u_N extends to $N=0$ as $u_0(p) = -p$, but the true mass of an unallocated prompt is 0 (as is a single-rollout group's, $u_1 \equiv 0$), so with optional activation ($N_i \geq 0$) the objective is not $\sum_i u_{N_i}(p_i)$ and the problem acquires an activation structure outside this proposition's scope.

Interpretation. A mass-optimal reference allocator for the static, known- p problem only; it does not prove that sampling prompts $\propto u^\gamma$ is optimal (at fixed group size, expected immediate mass is linear in the sampling distribution, so the unconstrained myopic optimum is a brittle arg-max), nor does it describe the §6.1 oracle, which samples prompts by true- p utility at fixed group size rather than varying N_i . It is also the idealized common form of across-prompt reallocation (our teacher) and within-prompt rollout adaptation (Reinforce-Ada): water-filling spends its next rollout wherever $p(1-p)^N$ is largest.

C Full details for compressed experiment rungs

Material moved from §6 during the page-budget pass; every number here is artifact-backed in the repository.

6.1 control battery (full). Eight arms, 5 seeds each: uniform 0.650, lr $\times$ 2 0.798, random-target relabels 0.800, live-gradient replay 0.832, true relabels 0.869, no-floor γ -matched oracle $0.8885 \pm .0014$, full stack $0.8895 \pm .0023$, oracle+recycling $0.8935 \pm .0012$. The relabel direction term (+.037 over replay) has seed spread $\pm .0025$, 4–6 $\times$ tighter than every placebo; its worst seed beats every placebo's best. Random-target relabels are harmful relative to dose-matched replay ($-.032$, 5/5 seeds) but neutral vs. plain lr-doubling: fake labels waste the slot rather than poison training. Utility-form controls: exact u_N $0.728 \pm .002$ vs. legacy frontier form $0.733 \pm .013$ (the derived form's payoff is 7 $\times$ smaller seed variance); the $N=2$ slice $p(1-p)$ is a wash against uniform ($-.007$, 2/5 wins) where u_N carries $+.078$ (5/5). AUC spread across arms is 9 $\times$ the final-performance spread.

6.2 full-control-variate baseline (full). The MaxRL paper's own Eq. (10) keeps the control-variate term $-\frac{1}{N} \sum_i S_i$ on all-fail groups, so that estimator is *not* silent at $K=0$ ($\mathbb{E}[\hat{g} \mid K=0] = \nabla p/(1-p)$, MC-verified). On the frontier-heavy pool it spends 100% of its update norm on all-fail groups and still scores 0.000 in every seed, with or without the teacher, while recycling under either variant reaches 0.98; on the balanced pool the two variants tie (paired Δ AUC straddles zero, 5 seeds). Artifact `results_fullcv_baseline.json`.

6.3 inference-efficiency detail. Convention: smallest k reaching absolute coverage 0.25, log-interpolated; artifact `maze_gpu/efficiency.json`; Fig. 10 plots the summed-coverage crossing. Summed coverage over L0–6: GRPO 3.24 vs. MaxRL 3.07 at pass@1; crossing at $k \approx 4$; 5.12 vs. 4.62 at $k=64$, with GRPO's level-2/4 curves saturating at 0.88/0.56 where MaxRL's reach 1.00/0.62.

6.5 IsaacLab protocol. Five arms: frontier teacher, stock greedy walker, uniform, scripted ramp, no-motion control; fixed-grid per-level evaluation; Anymal-C rough terrain; one seed, executed by an independent team against a co-designed, pre-registered protocol. The teacher's only edge over the greedy walker is a within-noise sliver at the hardest evaluated level.

6.6 gym control (full). MountainCar flag: measured random-policy success $< 3 \times 10^{-6}$ over 10^6 episodes; survive-400 $\approx 10^{-13}$ by tail fit. Dense-reward controls (prediction registered before the run): CartPole's native +1/step reward lets vanilla REINFORCE on the same tile policy solve survive-400 (0.96 ± 0.03 , 3 seeds); MountainCar's native -1 /step reward (constant until first success) stays at 0.000 in all seeds — the categorical zero tracks reward sparsity, not the environment. Attribution at a common 280-step horizon (hard-task AUC): MC .750 $\rightarrow$ +recycling .895 $\rightarrow$ +teacher .956; CartPole .708 $\rightarrow$.792 $\rightarrow$.893; paired teacher increment positive 3/3 seeds on both; creation-without-allocation plateaus at 0.958 on CartPole in every seed while adding the teacher restores 1.000; teacher speed effect 3.0 $\times$ to hard-task 0.9 (2.1 $\times$ to 0.99 on CartPole). The per-bin ablation replicates the 10-seed transition-matched study (flag 0.848 ± 0.058) before strengthening it to full convergence.

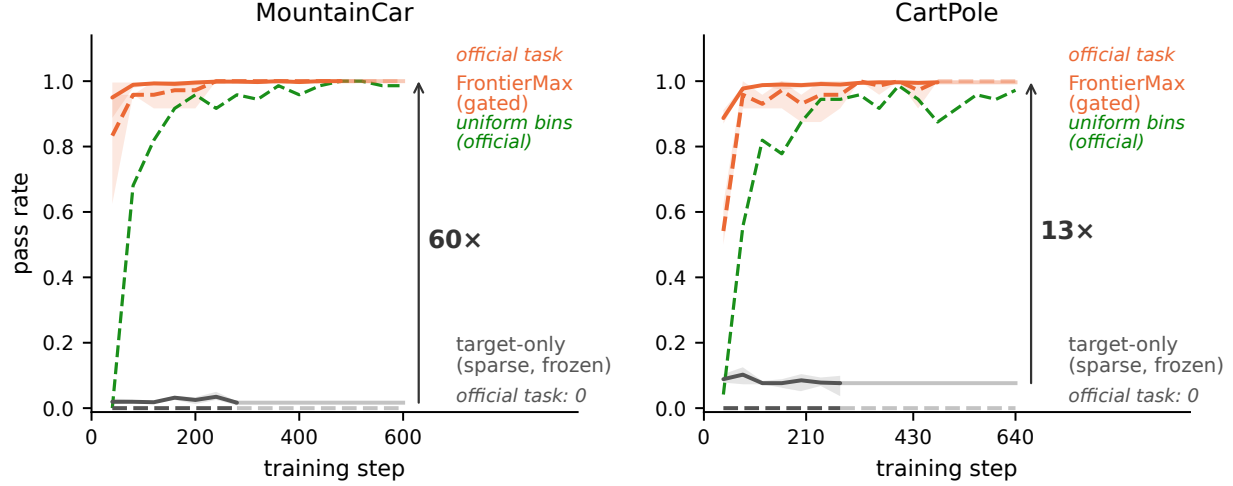


Figure 9: **Gym control under binary task-success reward, trained to plateau (3 seeds).** Solid: mean pass rate across curriculum bins; dashed: the *official* task (MountainCar flag / CartPole survive-400). The target-only arm (gray) is a provable zero-update freeze under sparse verifier-style reward; a uniform-over-bins control (green) also cracks the task, 3 $\times$ slower and off-ceiling; the gated FrontierMax stack converges to 1.000 ± 0.000 .

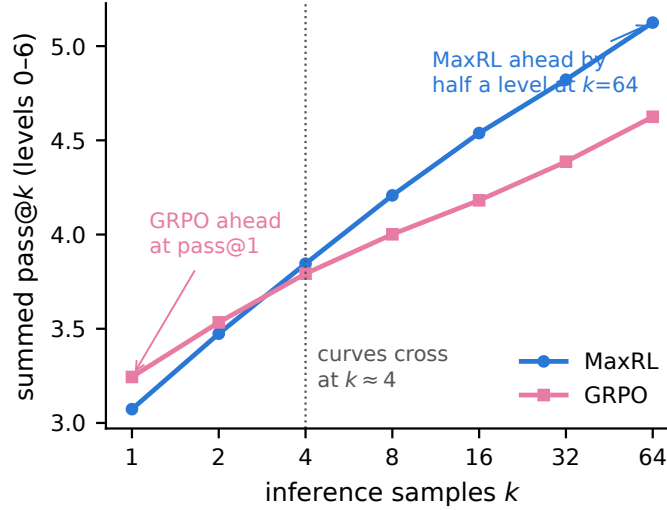


Figure 10: **Summed maze coverage vs. inference samples k (final uniform-arm checkpoints).** GRPO leads at pass@1; the curves cross at $k \approx 4$; MaxRL leads by half a level at $k=64$ — choose the estimator by how many samples you will draw at inference.

6.7 posterior diagnostics and harness caveats (full). The pass@ k sweep’s absolute level sits $\sim 3\times$ below the trainer’s val metric (unreconciled harness discrepancy; only the within-sweep contrast is quoted). Posterior detail: two independent teacher runs’ posteriors agree with each other (cross-run Spearman $\approx .45$ on co-visited prompts, $n \approx 900$; $\approx .67$ self-consistency with a shared latent) vs. $.10-.22$ against external 7B-model annotations, from ~ 1.25 visits per prompt. Across external-difficulty quintiles spanning solve rates $.33 \rightarrow .98$, mean $\hat{p}$ moves only $.076 \rightarrow .103$ — the compressed landscape that starves allocation. A response-length read we advanced after five runs (three GRPO runs compressed harder than both MaxRL runs) did *not* survive the sixth: the within-seed control run finished longer than every earlier run (252 vs. 208–236

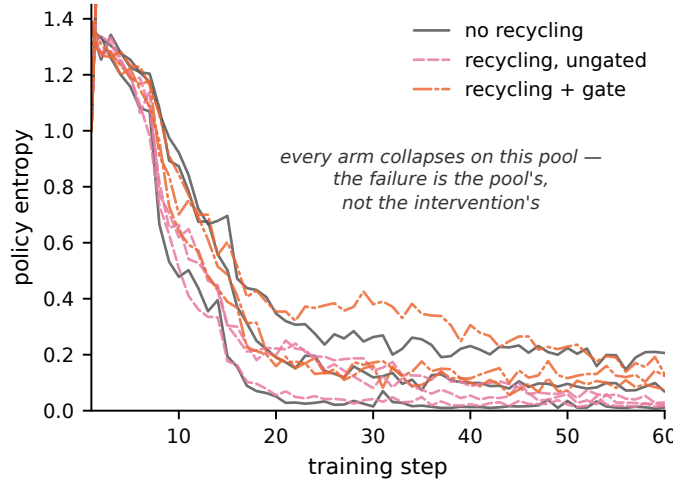


Figure 11: **Jugs pool-conditionality: every arm collapses (3 arms $\times$ 3 seeds).** On a pool whose learnable band is a single stratum of 1–2-move template solutions, plain MaxRL’s policy entropy collapses ($1.36 \rightarrow 0.01\text{--}0.21$) with no recycling involved; ungated recycling ends lowest, and the gate merely delays the fall. Rollout-set diversity at initialization is the design gate this suite motivates (§9).

tokens from a common ~ 365 -token warmstart), so output length does not separate the estimators at this scale and we withdraw it as a sharpening signature. Prompt-uniqueness figures for the replacement confound: $\sim 34\%$ unique prompts seen by teacher arms vs. $\sim 43\%$ uniform.

Relabeling in a PPO-family loop: the wiring contract. For adopters (the details a config cannot express). *Placement*: relabeling runs after reward computation and **before** `compute_log_prob` — old log-probs must be computed under the *rewritten* conditioning, making the relabeled group an on-policy ML term with the estimator’s own weights, not a policy-gradient ratio against stale context (our gridworld ablation priced the stale-context mistake at -0.06 AUC, worse than no hindsight). *Rewrite*: the prompt’s goal slot and the response’s goal mentions are both rewritten (token IDs, attention mask, and position IDs rebuilt; 2-D RoPE only — 3-D mrope is refused loudly); the answer artifact is untouched by construction. *Group semantics*: each parseable failure relabels to its own achieved goal and keeps its group id, so one group can mix relabeled tasks; unparseable rollouts stay failures, preserving contrast ($K < N$). The group contrast is then “reached some verified goal” vs. “reached none” — coarser than 1-of- N of a single task, pre-registered as a deviation, with the verl-normalized advantage verified equal to $N \times$ the reference weights in direction; as Remark 3 states, this per-row variant is a weighted-SFT update, not a per-destination MaxRL gradient. *Hygiene*: the curriculum posterior observes pre-relabel scores only; no KL-to- reference is applied to relabeled groups in our runs (`kl_coef=0`). Reference implementation vendored in this repository (`verl_integration/vendored/hindsight.py`, ~ 270 lines, with execution-commit provenance and CPU integration tests).

Jugs suite (pool-conditionality; §9). Tier grid t0–t4 (2–5 jugs, disjoint min-moves bands, 200 unique tasks/tier); feasibility landscape (SmolLM2-360M two-shot, $N=16$): t0 .058/.475 (pass@1/pass@16), t1 .006/.100, t2–t4 0/0 with relabel-yield-on-failure .62–.73 on every tier. All nine pre-registered B-arm cells (no-recycling / ungated / gated $\times$ 3 seeds) converged to the same deterministic fixed point: t0 mean@16 = pass@16 = 0.450 exactly (the 2-move template stratum), t1 pass@16 $0.15 \rightarrow 0$, actor entropy $1.36 \rightarrow 0.02\text{--}0.21$. Ungated recycling ran 169 relabels/step at yield .88 and ended with the lowest entropy (0.02–0.03); the value-keyed gate rejected 99.8% of relabels (the granularity bug described in §9, since fixed and regression-tested). Fig. 11 shows all nine trajectories: the collapse is the pool’s property, not any intervention’s. Postmortem and per-cell artifacts in the repository.

6.9 secondary reads (full). Pool construction and baseline calibration: the pre-SFT base model’s re-label yield is 0.000 ($n=384$) — the syntactic prerequisite; post-SFT the model is a near-independent guesser whose pass@16 sits at a tier-invariant $\sim 75\%$ of the independence bound $1 - (1 - \text{pass}@1)^{16}$ (ratios .744/.783/.755 across tiers), the baseline the headroom gate measures against. Training-side damage: ungated recycling ends with more unsolvable-at-16 frontier training prompts than baseline in every seed (paired +.083/+.063/+.034). Entropy at 3 seeds: the gate’s retention replicates (+.072 paired over baseline, all seeds positive, $\sim 1.6\times$); ungated recycling’s endpoint entropy is seed-unstable (spread $\pm .062$, $4\text{--}9\times$ the other arms’), consistent with its noisy frontier-coverage endpoints. Designed-strength sweep (P-R1) per-seed tier-1 endpoints (mean@16/pass@16, rejection fraction): .212/.513 (.934), .322/.573 (.934), .264/.488 (.944) — the registered windows were mean-kept $\in [0, .60]$ of B2’s gain and coverage $\in [.541, .571]$; observed mean-kept -0.26 , coverage .525. Artifacts `armA_b3fix_s{1,2,3}.json` with prereg and verdict in `countdown_reviewer_arms/`.

D Claims and evidence

Every load-bearing claim, its strongest evidence, and its pre-registered status. “Registered” means the prediction and its falsification branch were committed before the deciding run; executed branches are marked. Artifacts named per row live in the repository with the run registry (`run_registry.json`).

E Hyperparameters and compute

Total compute for every experiment in this paper: ~ 230 A10G-hours on a single GPU plus CPU-days for the exact-gradient suites. Step-time telemetry (artifact `generation_timing.json`, 1,188 mixed GSM8K/Countdown steps): step-weighted mean generation fraction 31% overall; the eight generation-dominated sessions ($>80\%$ session mean, 282 steps) average 83% — a selected-subset description, not an estimate of typical cost. Table 3 gives the per-suite configuration and Table 4 the knob inventory with provenance.

Reproducibility

One command verifies the artifact end to end: `bash reproduce.sh` runs the proposition and integration tests, re-derives the figure endpoints from per-seed artifacts (failing on mismatch), checks every manifest checksum, regenerates every figure from its frozen inputs, and spot-checks each quoted prereg verdict against its committed artifact (`--build` additionally rebuilds both PDF wrappers). Every proposition has an enumeration/Monte-Carlo verification script (`test_mass_formulas.py` in `curriculum_maxrl/` checks the MaxRL, RLOO, and both GRPO-SD mass formulas against exact enumeration and the deployed code). Every figure regenerates from committed inputs listed, with checksums and commands, in the frozen manifest `paper/results/manifest.json`: empirical result values are declared as versioned structured inputs under `paper/figures/data/`; the former hard-coded seed-1 Countdown curve now loads its checksummed bootstrap-proxy table, and the maze cohort of Fig. 5 is frozen in an explicit run manifest — adding a run file cannot silently change the figure, and missing inputs are a hard error. The Jugs preregistration, raw cells, verdicts, and postmortem are vendored in `jugs_llm/results/` with preregistration- and execution-commit provenance. The LLM-scale relabeling implementation is vendored verbatim with execution-commit provenance (`verl_integration/vendored/`), so the loop the Countdown and Jugs suites executed is inspectable in this repository. Known gaps, stated rather than papered over: the IsaacLab pilot is reported from an independent team’s summary (raw logs live with that team); per-step timing fractions come from verl’s in-run telemetry (`generation_timing.json`); the fig2/fig3 endpoint tables were transcribed from run analyses, and all of fig2’s quantitative panels now carry checked-in derivation scripts that regenerate them from per-seed artifacts and fail on mismatch (`verify_fig2a_from_artifacts.py` covers panels a–b, `verify_fig2c_from_logs.py` panel c against raw maze seed logs) — only the GSM8K panel (d) and fig3 remain transcribed, with per-panel provenance in the data files; and the GSM8K pass@ k harness sits $\sim 3\times$ below the trainer’s val metric (unreconciled; only within-sweep contrasts are quoted, and the reconciliation script is queued behind the running

Table 2: Claim inventory. $\uparrow$ = confirmed as registered; $\downarrow$ = refuted, committed branch executed; $\circ$ = descriptive/exploratory (no registered endpoint); $\rightarrow$ = registered, run pending.

claim	strongest evidence	status	where
mass identity $A_N = 2u_N$; factorization $\mathbb{E}[\hat{g} x] = u_N(\mu_+ - \mu_-)$	proof + MC (<code>verify_guidance_math.json</code>); enumeration test in CI	proved	Prop. 1
drop- $K=0$ estimator unbiased for $T=N-1$	three-line derivation + SymPy check	proved	Lem. 1
allocation saturates: oracle ties full stack creation is the only live channel at $p \approx 0$	no-floor γ -matched oracle .8885 $\approx$.8895, 5 seeds	$\circ$	§6.1
	uniform+recycling .931 $\approx$ teacher+recycling .928 vs. 0.000 all pure samplers, 5 seeds	$\circ$	§6.2
zero-exception coverage divergence (9-run cohort)	failed balanced factorial at registered endpoint (3/6, 4/6 vs. $\geq 5/6$ both)	$\downarrow$ retracted	§6.3b
time-integrated coverage ordering (MaxRL > GRPO)	wave-2 registered primary on fresh blocks: 6/6 per sampler, $p=.031$ each; 12/12 independent blocks across waves	$\uparrow$	§6.3b
easy-band concentration of GRPO's loss	wave-2 registered P-F3: pair-level bar met at 10/12; block-level 4 positive, 1 tie, 1 negative, CI includes zero	suggestive	§6.4
variance normalization is the tail mechanism	exact rung: no-SD GRPO collapses onto RLOO (.148 vs. .161); neural rung: P-G0c <i>failed</i> (loses easy band 6/6)	split	§6.3b–6.4
GRPO's own-mass teacher doesn't rescue it	P-G0a confirmed 5/6, both rungs	$\uparrow$	§6.3b
exact-rung ordering survives lr recalibration	P-LR1 falsified as registered: 2–4 $\times$ lr closes raw Δ cov but premium collapses (.007–.013 vs. .050) — restated at-matched-lr	$\downarrow$	§6.3b
recycling-package sharpening (mean up, bootstrap proxy down)	reported 3-seed aggregate; full k curve is exploratory seed 1; per-task outcomes absent	aggregate	§6.8
higher-dose extra-updates bound at LLM scale	replay (2 epochs) beats recycling's mean gain <i>with</i> coverage gain, P-R2 3/3 seeds; does not isolate direction	bounded	§6.8
moderate under-gated point	frontier-tier coverage to baseline at $\sim 60\%$ mean kept under buggy decay; corrected validation open	provisional	§6.9
gate strength is a monotone dial	designed-strength sweep P-R1: mean-kept $-0.26 \notin [0, .60]$, coverage .525 < .541, 3 seeds	$\downarrow$	§6.9
estimated- p gate is affordable	P-PB1 confirmed: one probe rollout/step recovers 98% of the oracle-vs-frequency gap at matched total rollouts (10/10)	$\uparrow$	§6.9
LLM teacher $\times$ estimator interaction (GSM8K)	original shape 1-of-2 seeds; strengthened arm failed preregistered delivery gate	inconclusive	§6.7
pool-conditionality: Jugs defeats every arm	pre-registered all-null with entropy-collapse mechanism	$\uparrow$	§9

arms). A one-row-per-run registry (`curriculum_maxrl/run_registry.json`; 53 rows spanning the maze, Countdown, and GSM8K artifacts, with arm/seed/protocol per row) indexes every run this paper draws on; the maze main-effect subset and its 5v4 composition are tabulated in `permutation_reanalysis.json`, committed alongside. Data hygiene is checked, not assumed: both RL pools carry zero train/eval task overlap and zero intra-split duplicate tasks, but the historical Countdown SFT warmstart overlaps 27/128 tier-0 evaluation tasks; tiers 1–2, which carry the central sharpening claim, have zero SFT overlap. Tier-0 absolute values are therefore contaminated and excluded from clean absolute claims. The rebuilt matched-control split has zero SFT/test overlap in every tier (Countdown identity = target plus sorted operand multiset; GSM8K = prompt text; integrity artifacts accompany each split). The framework is a dependency-light package (`frontier_rl`, numpy-only core) with adapters for LLM pools (verl), gym control, IsaacLab parallel sim, and flow-policy VLAs.

Table 3: Per-suite configuration. Teacher knobs (decay, floor, γ) are shared across suites with the defaults below; what varies per suite is scale, not schedule.

	chains (§6.1–6.2)	maze (§6.3–6.4)	GSM8K (§6.7)	Countdown (§6.8–6.9)
policy	exact tabular	1.26M transformer	SmolLM2-360M	SmolLM2-360M (SFT)
rollouts N	16	32	16	16
groups/step	1	8	64	64
optimizer, lr	SGD, 0.5	AdamW, 10^{-4}	AdamW, 10^{-5}	AdamW, 10^{-5}
budget	3,200 groups	2,400 s / 250 steps [†]	50 steps	60 steps
seeds (headline)	5–10	3 / 6 blocks [†]	1–2/cell	3/arm
compute/run	CPU-min	0.7 A10G-h	~20 A10G-h	~6 A10G-h

[†]cohort runs (§6.3a) used 2,400 s wall-clock; the balanced factorial and its confirmation wave use 250 fixed steps (matched generation and optimization budget) over six seed blocks — the two protocols are never mixed within a comparison.

Table 4: Teacher and recycler knobs: value, provenance, and measured sensitivity.

knob	default	derived or tuned	sensitivity
score shape / N -scaling	—	derived (Prop. 1)	shape supported; peak <i>location</i> rejected (§8)
posterior decay	0.7	tuned (V2b)	closes ~19% of oracle gap
uniform floor	0.1	tuned	0.15 on maze; robust 0.05–0.2
γ (concentration)	1	tuned (V6)	4 on chains; no transfer to maze
<code>gate_max_p</code>	0.5	tuned ($p=1$ zero vacuous)	corrected strong setting failed (§6.9)
relabel dose cap	12 groups/step	tuned	dose effect remains descriptive (§6.4)